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AN OPTIMIZATION APPROACH TO FANTASY FOOTBALL DRAFT AND ROSTER MANAGEMENT

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ABSTRACT

Fantasy football is already one of the most popular games in the United States, and its player base only grows each year. The fantasy sports industry is worth billions, and this is in no small part to the game's parallels to gambling—each fantasy player contributes money to be re-distributed based on a fantasy league's final standings; in doing so, the players enter into a game of risk and expected return. The goal of this applied research is to look at techniques in two primary phases of the fantasy football season that can assist a fantasy player in making sound decisions.

First, the research considers decisions made during the initial draft selection process of a fantasy football league. This piece of the research considers the tradeoff between the projected score of an NFL player and their demonstrated historical weekly variance in performance. This comparison of projected scoring and demonstrated variance is carried out using a bi-criteria optimization model. Considering the draft behaviors of opponent owners, simulations were carried out to assess the effectiveness of the mean-variance comparison in drafting fantasy teams. An efficient frontier of teams was created to demonstrate the tradeoff between expected score and demonstrated variance, and a set of teams from that efficient frontier were compared. As a result of the analysis, it was found that modeling player selection at the draft stage as an effort to maximize projected scoring and minimize total team variance could result in a team of players which scored near the maximum projected number of points with a substantially lower variance.

The second part of this research considers season-long ramifications of roster changes and starting lineup selections made at single decision points throughout the course of the season. First, a model was created to maximize the total number of points scored by players in the starting lineups over the course of the season. Next, an algorithm was created using this maximization

model to test the "value added" from making an "add/drop" move in free agency. In other words, the effects of a single-week roster change were assessed based on the move's effects on the team's scoring projection for the entire season. This novel consideration of the season-long impact of weekly moves provides a formal in-season decision-making method that projects to only improve the outlook of a fantasy team.

Finally, the draft optimization technique and free agency assessment algorithm were considered in concert. A data-drive optimization model for the draft provides a fantasy team owner with a consistently high scoring team and model which is robust to the behavior of other fantasy owners at the draft. Additionally, a simple maximization technique allows the team owner to project their fantasy team's final point output based on optimal starting lineups. Finally, at each week throughout the regular season, the team owner is given the ability through the free agency optimization algorithm to assess the long-term impacts of any immediate moves that may be under consideration. In aggregate, these techniques combine to produce fantasy football teams with high season scores and low variances which prove to be adaptable and robust throughout the season.

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Chapter 1

Introduction

The history and scope of fantasy football and the primary objectives of this research are discussed in this chapter.

1.1 History and Scope of Fantasy Football

Fantasy football is one of the most widely played games in the United States, and its following has only been growing in recent years. Started in 1962 by a part-owner of the Oakland Raiders football team (Augustyn & Zegura, 2016), fantasy football involves a set of league "owners" who "draft" actual National Football League (NFL) players to their fantasy teams and accrue points each week based on the actual performance of those players. Back then, the group dubbed themselves the "Greater Oakland Professional Pigskin Prognosticators League," but today, they're known as the founders of a multi-billion-dollar hobby held by millions around the world.

Today, more than 59 million Canadians and Americans play fantasy sports, and over 80% of those people play fantasy football (McCormick, 2019). With the number of fantasy players rising year-over-year, a 2017 evaluation that the fantasy sports industry was worth $7 billion a year may at this point be a significant under-evaluation (Voice of America English News, 2019). The highest winning fantasy sports bettors understand statistics and can apply it to this type of format; at a minimum, the most successful group of players have ratings, models, and Excel trackers to aid them in making decisions (Baer, 2015). Fantasy football competition has advanced significantly since it began as a group of friends in 1962; today, it's a money-making behemoth, and those with statistical chops are the ones coming out on top.

1.2 The Basic Rules of Fantasy Football

Typically, fantasy football leagues consist of between 10 and 16 teams, and each team is made up of between 12 and 16 actual NFL players. These rosters are first filled during an initial draft—an event before the regular NFL season begins during which team owners (TO) choose NFL players to add to their team. While there are multiple methods for conducting a fantasy football draft, each accomplishes the same goal of splitting up NFL players between fantasy teams and creating initial rosters for each team owner.

Each fantasy team is made up of a specific number of players from each position. Most fantasy leagues dictate that each team may play at most one quarterback (QB), two running backs (RB), two wide receivers (WR), one tight-end (TE), and one "flex player" (FLEX), who may be a running back, wide receiver, or tight-end; additionally, each team also may play up to one defense/special teams unit (D/ST) and one kicker (K). This group of players makes up a roster's starting lineup, and only these players accrue points for a fantasy team during a week; the number of fantasy points scored by a player is based on their actual, in-game performance. In addition to the players in a team's starting lineup, the fantasy team owner may also draft a number of players to include on their bench; in other words, these players do not accrue points for the team each week but are available to insert into the starting lineup should they face a more advantageous matchup or a regular starting player suffer an injury. In the event that a player underperforms for a prolonged period or becomes injured with little chance of returning to play, a TO may even choose to "drop" that player and “pick up” another player as long as that player is not already on another fantasy team.

Each NFL season is made up of 17 regular-season weeks. Teams play 16 games over that stretch, with one week designated as the team’s bye week in which they do not participate in a game. Fantasy football seasons typically last about 16 weeks, with 13 of those devoted to regular season play and the final 3 devoted to a tournament-style playoff. During each week, two fantasy teams go head-to-head, with each team’s starting lineup accumulating points. At the end of the week, the fantasy team that has accrued the most points is declared the winner of that contest, and the league progresses to the next week, resetting all fantasy scores and changing head-to-head opponents. Based on regular season win-loss records, fantasy teams are pitched against each other for a playoff tournament, and the winner of that tournament is declared the winner of the fantasy football league. In nearly every league, there is some type of monetary payout to the top players; in some leagues, the league winner may take all of the prize money, while in others the winner may take 50% of the winnings while the runner-up could take 30% and the third-place player could take the final 20%. In any case, fantasy football success typically carries a significant financial incentive.

1.3 Objectives

The goal of a rational football player is to win the most money possible based on the prize structure of that player’s fantasy league. To do this, the TO’s goal should be to win as many games as possible; because this objective has built-in uncertainty due to the uncertain nature of opponent scoring, a TO should essentially aim to maximize the total number of points scored by their team while minimizing the week-to-week variability of their team's scoring. Scoring highly and consistently each week should give a TO the best chance to maximize the number of wins accrued by their team. While many team owners choose the players to add to their teams based on expert rankings and projections, the best players employ models, ratings, and trackers. With all of this in mind, our objective for this research is to create a robust strategy for choosing and maintaining the best possible fantasy football team.

In terms of team-building, the fantasy football season can be broken up into two periods: the draft period, and the in-season period. The draft is a single event during which NFL players are drafted by team owners and put on fantasy teams. This is the most significant point of player assignment during the fantasy contest, and it sets up each team with a set of primary resources to be used in winning games the rest of the year. As ESPN's Senior Fantasy Sports Analyst, Matthew Berry, often says, though, "You can't win your league at the draft, but you can lose it" (Berry, 2019). You can't win your league at the draft because a significant amount of player movement and re-assignment and fantasy player decision-making comes after the initial draft phase.

During the season, team owners have two major considerations regarding the players on their teams: 1) Should they drop a player in favor of adding another one to their team, or trade a player for a different one, and 2) Should they insert a player from their bench into their starting lineup? These week-to-week decisions are what makes or breaks a successful fantasy season. Adding a player to your team early in the year and having them become an integral part of your team's success the rest of the season can provide a huge boost to your chances of winning your fantasy league. Likewise, choosing to play one wide receiver from your bench with an easy matchup instead of another wide receiver from your starting lineup with a difficult matchup could prove to be the deciding factor in a weekly contest if that typically weaker receiver contributes a high score. While many players consider statistics during a fantasy draft, most fail to use data to drive their in-season decisions, despite the fact that these in-season decisions are truly the ones that set a fantasy team up for success.

Because of the split-period nature of a fantasy football season, the objective of choosing and maintaining the best fantasy football team must be broken up into two parts. First, our objective is to create an optimization model and algorithm to be used during a fantasy draft through which we can select a fantasy team of players with the highest expected number of points scored and lowest week-to-week variability. This also necessarily entails predicting the behavior and draft strategy of other fantasy team owners.

Second, our in-season objective is to create a technique to assess whether a fantasy team owner should drop one of its current players so that they can add another player instead and to maximize the total number of points scored by starting lineups over the course of the season. This dual objective is accomplished by first creating an optimization model to choose the players from a roster that should be included in a fantasy team's starting lineup during a given week. Second, an algorithm is created to assess the "value added" contribution of free agent players, leaving the team owner with suggestions for adding players from free agency and dropping players from their roster. In combination, these techniques designate the set of players most likely to accrue the highest number of points for the team over the course of the season.

In all, these objectives combine to create a robust season-long strategy for drafting and maintaining the team most likely to win a fantasy football league. Each of the following chapters has a different scope. Chapter 2 reviews previous literature published on the subject of fantasy team optimization and fantasy sports betting. Chapter 3 reviews the datasets available for use in our research. Chapter 4 proposes an optimization model and algorithm to be used during a fantasy football draft to choose an optimal team. Chapter 5 addresses in-season dynamics, with proposals for optimization models surrounding adding and dropping players and selecting players to insert into a starting lineup. Finally, Chapter 6 provides a set of conclusions gleaned from the research as well as potential areas of future further study.

Chapter 2

Literature Review

This is certainly not the first attempt to marry fantasy sports with robust statistical analysis or optimization modeling. Most publications in the realm, though, focus specifically on the gambling implications within the confines of daily fantasy sports contests. Many papers focus on player selection methods with the objective of maximizing the chances of creating a daily fantasy team that scores well enough to win money. The other application of similar techniques to fantasy sports has been the prediction of player statistics—specifically the prediction of player statistics as a result of previous performances and future opponents. While most fantasy sports players rely on predictions from host sites such as ESPN and Yahoo, novel approaches to predicting NFL player performance have yielded accurate predictions as well. With all of that said, very little has been written specifically addressing the application of optimization modeling techniques to season-long fantasy football play, though a review of relevant literature provides some points of consideration in developing an optimization methodology.

The thesis published by Paul Martin Andrejko (2011) is the first relevant academic review of one of the core pieces of choosing a winning fantasy team: player projections. In the paper, Andrejko compares the player output predictions of three host sites with the actual player output, also considering and analyzing the differences in accuracy of predictions for players at varying positions. Andrejko concluded that there was a statistically significant difference between the accuracy of predictions from different host sites and also found a significant difference in the accuracy of predictions for players at different positions; these conclusions were made by computing the squared error score based on player rank. Unfortunately, this was a relatively limited analysis, looking only at three host sites and two positions.

Paul Steenkiste (2015) took steps towards modeling player performance. The goal of the research was to predict weekly player performance based on the previous few weeks of performance; notably, Steenkiste looked at individual statistics as opposed to a simple singular metric of fantasy points scored. While the modeling techniques used in the paper—linear regression, Random Forests, Multivariate Adaptive Regression Splines (MARS), and weighted MARS—performed well in comparison to predictions from host sites, the ultimate takeaway was the predicting individual statistics for single players in single weeks can be a particularly futile practice.

The most recent attempt at applying true modeling practices to fantasy draft procedures comes from Haugh and Singal (2019). While the focus of their research is daily fantasy sports, the models proposed in the paper maximize the expected reward of money won in the league subject to portfolio feasibility constraints taken from daily fantasy sports rules and roster regulations. Essentially, the model focused on maximizing the portfolio mean score awhile constraining asset variances and inter-portfolio correlations. The research only uses data from one NFL season and poses questions regarding other sports and effective methods of modeling opponent portfolio selection.

Perhaps the most applicable piece of previous research is the work of Sun and Becker (2016). The aim of their research was to build a mixed-integer optimization model using predictions for draft selection and weekly lineup management with the objective of winning a fantasy football league. Their draft selection methodology primarily used predicted points scored by week, although their prediction method proved to over-project player weekly output by an average of 4.5 points per player per week. Additionally, Sun and Becker do not take variability in player performance into account, although they are the first to effectively address the issues of weekly play—a simple greedy algorithm was employed to select starting lineups with the highest projected points by position. These modeling techniques yielded good results, showing that the probability of finishing in the top 2 places of the fantasy league was significant at a 99.75% confidence level based on 300 simulations.

While Sun and Becker's paper describes a method that is successful in improving the average final placement of a fantasy sports team, their method leaves much to be desired. Their draft algorithm uses weekly projections, which are inaccurate for weeks later in the season. Additionally, the objective function of the model includes weighted summations of points scored in the regular season, points scored in the playoff period, and expected number of wins. Understanding that injuries, role changes, team changes, and more can significantly alter the projected output of NFL players, this objective function heavily weights performance in weeks with inaccurate projections.

Sun and Becker's model also does not consider the variance of player performance at any point and uses only a simple greedy algorithm for weekly lineup selection. On the other hand, the model put forward by Haugh and Singal uses mean-variance optimization techniques in weekly lineup selection, although their model is not directly applicable to the same problem since it focuses on weekly fantasy contests.

Sun and Becker as well as Haugh and Singal address the question of modeling opponent behavior as an important piece of player selection. Haugh and Singal use Dirichlet-multinomial data generating processes for modeling opponent team selections with estimated parameters based on Dirichlet regressions. Sun and Becker, on the other hand, uses a weighted uncertainty set based on widely available player rankings to predict opponent behavior. Again, these methods are not directly comparable, as Haugh and Singal analyze daily fantasy sports contests and Sun and Becker address the question of full-season play split up into a draft phase and weekly play phase.

Fry, Lundberg, and Ohlmann (2007) model the decision-making process of a single real-life sports franchise during a player selection draft. Using a stochastic dynamic program with additional assumptions to create a tractable deterministic dynamic program, Fry, Lundberg, and Ohlmann effectively determine a player's value added to a team by distinguishing whether that player is anticipated to start or act as a reserve. By modeling the draft as a series of decision epochs at which the decision maker must select, the decision maker can take into consideration a player's estimated value, the availability and value of other players, and the team's need at a position. Applying a similar tactic to a draft optimization model for a fantasy team could prove to be valuable in selecting players who are most likely to add value (i.e. expected points) to the fantasy team.

Based on these papers, a few clear takeaways are evident. First, as in Sun and Becker's model, optimization phases should be split up into a draft phase and a weekly play phase. While the weekly play phase optimization algorithm used in Sun and Becker's model was only a simple greedy algorithm, observing the tactics uses in daily fantasy sports models, such as those of Haugh and Singal, may provide better weekly play phase performances. Additionally, as described by Fry, Lundberg, and Ohlmann, dynamic programming can be used in considering moves which may be made to supplement a roster. Either in the draft phase or the weekly play phase, the addition of players to a team at the expense of dropping an existing team player can be weighed by modeling the problem as a dynamic programming problem. Finally, modeling opponent behavior is essential for creating a draft optimization model. The best performing optimization programs, as shown in Sun and Becker, Haugh and Singal, and Fry, Lundberg, and Ohlmann, each consider the behavior of opponents in drafting players.

Chapter 3

Data Analysis

3.1 League Description

Fantasy football leagues are based on the performance of actual NFL players. Considering quarterbacks, running backs, wide receivers, tight ends, kickers, and defenses/special teams, there are well over 300 players available to fantasy team owners. In a standard ESPN league with 12 teams, up to 192 players can be held on rosters at a given time (ESPN, 2017). Including injuries to players, NFL teams signing free agents, and NFL teams promoting players from their practice squads, there is a huge number of players available for team owners to add to their team, and that list of players changes constantly throughout the season.

3.2 Dataset Descriptions

3.2.1 Data Sources

In order to retrieve all of the historical statistics necessary to complete this analysis, data was taken from a number of sources.

SportsDataIO.com’s (2020) Data Dictionary provided most historical fantasy football data. Season projections for players and weekly projections for players for the 2016, 2017, 2018, and 2019 seasons were taken from this site.

AdvancedSportsAnalytics.com's (2020) NFL raw data provided week-by-week scoring for each player for the 2016, 2017, 2018, and 2019 seasons.

Finally, FantasyPros.com (2020) provided historical data on the average draft position for all players in fantasy drafts for the 2016, 2017, 2018, and 2019 seasons.

Aggregated, these three data sources provided all of the data points required to carry out this analysis and modeling procedure. ESPN.com's fantasy section provided guidelines pertaining to fantasy team rosters and player limits.

3.3 Fantasy Team Description

Each fantasy football host site dictates a typical makeup of a fantasy football roster. These guidelines lay out the number of players from each position that may be included in a starting lineup as well as the total number of "bench" spots reserved for players held on the team but not included in the starting lineup. Additionally, these guidelines outline upper and lower bounds on the number of players of each position that may be held on any one fantasy team. For the purpose of simplicity, we'll adopt ESPN's standard football scoring and roster settings in considering roster constraints and scoring (ESPN, 2008).

ESPN standard league roster have 16 total players per fantasy team. Nine of those players are placed into the starting lineup, and seven are held on the bench. Bench players are unable to accumulate points for their fantasy teams, while the performances of starting players sum to the total score of the fantasy team for each weekly contest. At the conclusion of each week, players may be moved between the starting lineup and bench spots, allowing different players to accrue points in different weeks. Table 1 below shows the roster breakdown of an ESPN standard fantasy football team, with position requirements for the starting lineup and bench.

Table 1. Fantasy Football Team Roster Breakdown

Player Position	***Number of Players***
QB (Starting)	1
RB (Starting)	2
WR (Starting)	2
TE (Starting)	1
FLEX – RB/WR/TE (Starting)	1
D/ST (Starting	1
K (Starting)	1
Bench	7

Table 2 below shows the roster requirements dictated by ESPN's standard fantasy league rules. Of course, the roster breakdown shown above in Table 1 provides some guidance as to how roster spots can be allocated to different positions, but ESPN also places a cap on the number of players at each position that may be held on any one team, including bench spots. For instance, any one fantasy team may roster no more than six quarterbacks (QBs); indeed, if a fantasy team owner held one quarterback in their starting lineup and four others on their bench, they would need to drop a quarterback to free agency before adding another to their team roster. In combination, the roster breakdowns by position from Table 1 and Table 2 provide minimum and maximum position requirements respectively for a fantasy team roster.

Table 2. Fantasy Football Team Roster Requirements

Player Position	***Maximum Number on Roster***
QB	6
RB	8
WR	10
TE	5
D/ST	4
K	5

In order to consider only players who would realistically contribute to a fantasy team, we need to cut down the number of players from the NFL universe which we consider for addition to the team owner's fantasy roster. While each NFL team likely holds three quarterbacks on its roster, there are almost no cases in which any quarterback other than the top one on that team's depth chart would be relevant for fantasy purposes. Similarly, each NFL team likely holds three or four running backs on its roster, but there are almost no situations in which a running back outside of the top two on their own team would be relevant for fantasy analysis. After analyzing player performance data and observing steep performance cut-offs after a certain number of players at each position, the universe of players that we will consider for the purpose of this exercise is shown in Table 3 below.

Table 3. Player Universe Under Consideration for Modeling

Player Position	***Number of Players Considered (By Ranking)***
QB	18
RB	60
WR	75
TE	25
D/ST	11
K	25
TOTAL	**214**

3.4 Description of Distributions of Model Parameters

In order to make sound decisions in both the draft and weekly play phases of the fantasy football seasons, certain parameters must be estimated based on the past performance of players. For the player limits dictated above in Table 3, these parameters were estimated based on the 2016 performance of players.

3.4.1 Season Fantasy Parameters

Beginning with season fantasy scoring and projections, we can look at each position to observe the distribution of projected season fantasy scoring and actual fantasy scoring for the top number of players, as dictated in Table 3.

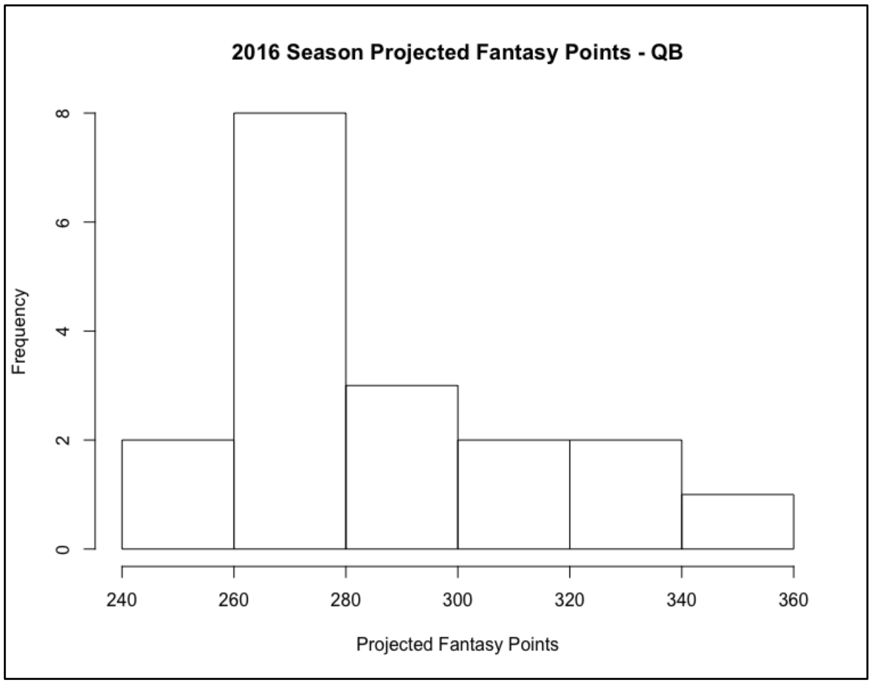

Figure 1. 2016 Season Projected Fantasy Points – QB

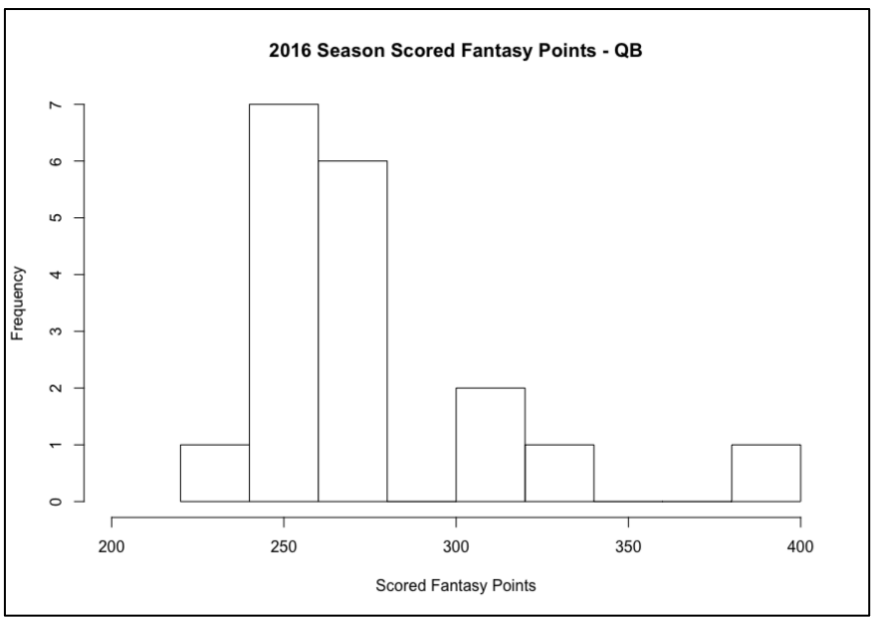

Figure 2. 2016 Season Scored Fantasy Points - QB

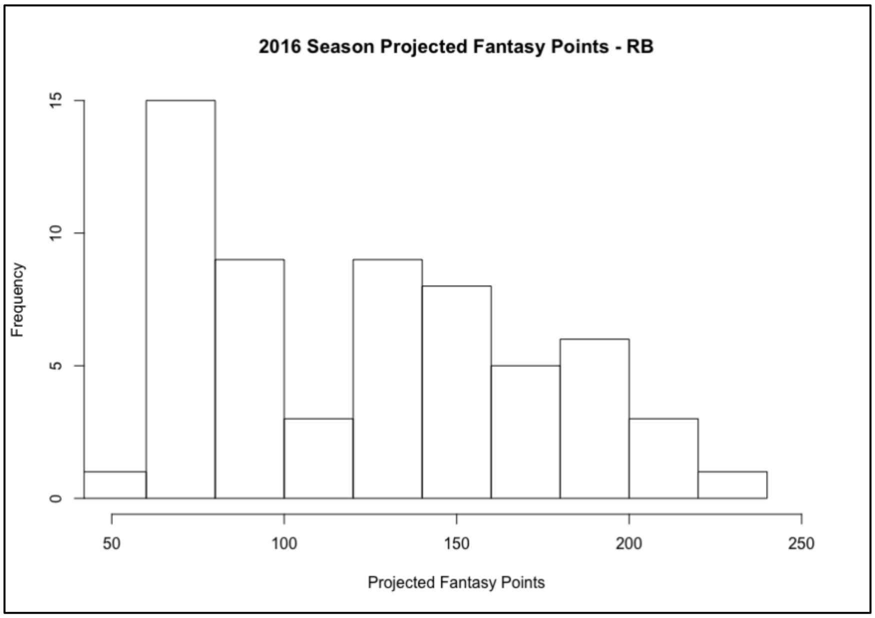

Figure 3. 2016 Season Projected Fantasy Points - RB

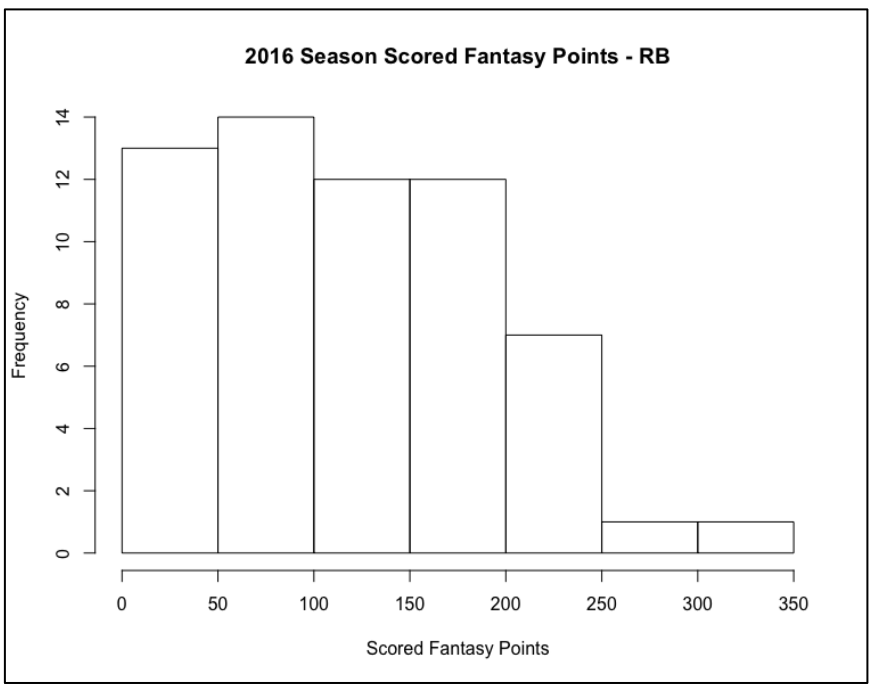

Figure 4. 2016 Season Fantasy Points Scored – RB

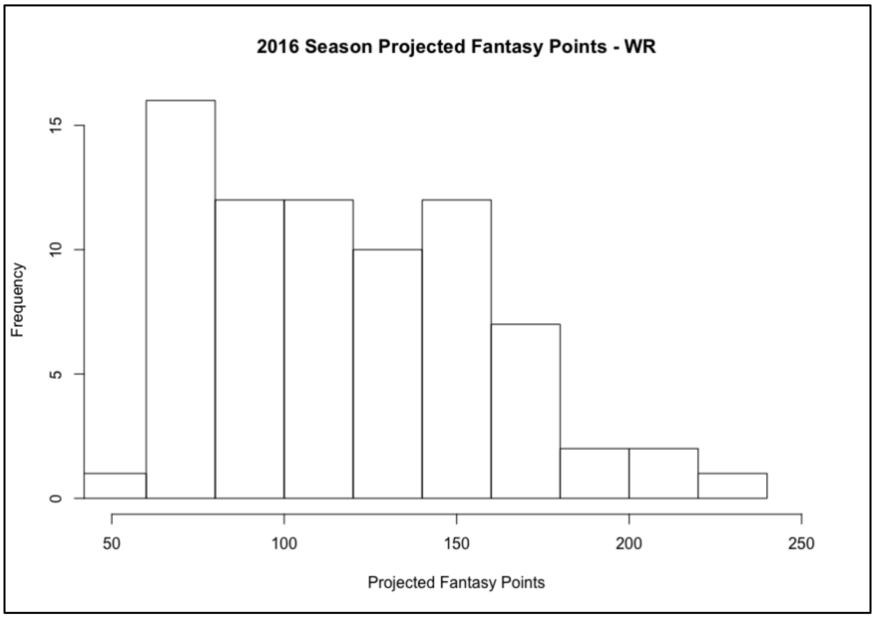

Figure 5. 2016 Season Projected Fantasy Points – WR

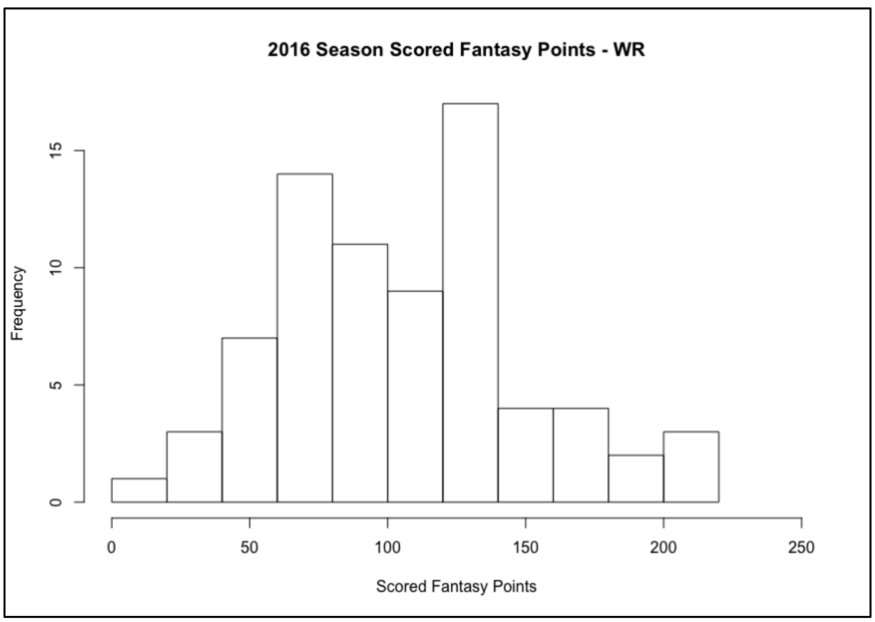

Figure 6. 2016 Scored Fantasy Points - WR

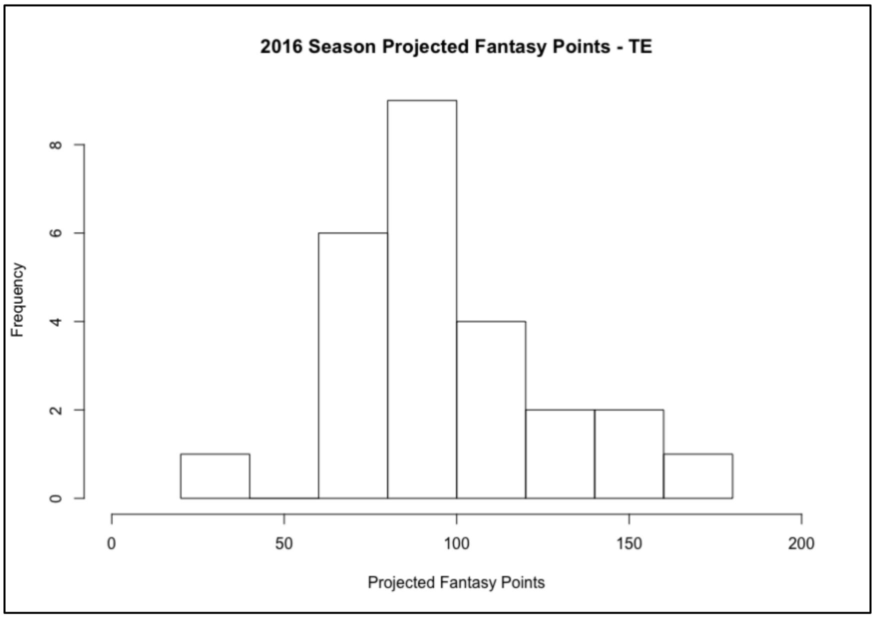

Figure 7. 2016 Season Projected Fantasy Points – TE

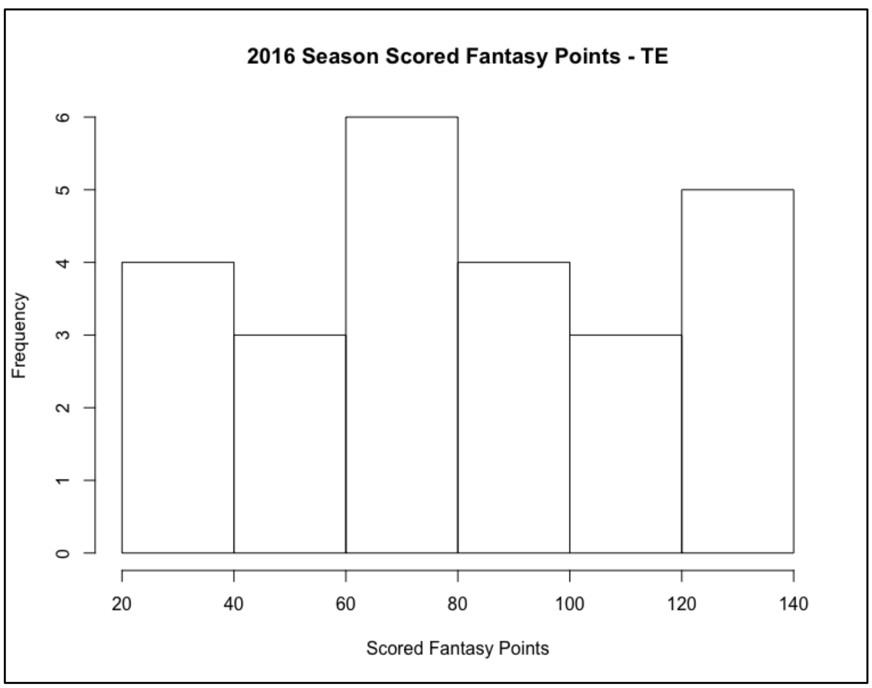

Figure 8. 2016 Season Scored Fantasy Points – TE

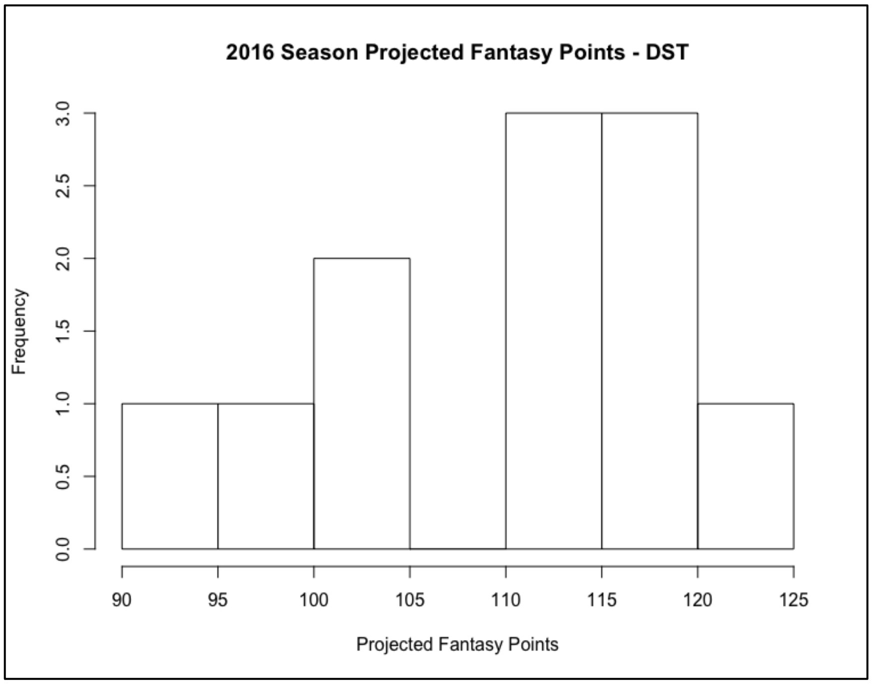

Figure 9. 2016 Season Projected Fantasy Points - D/ST

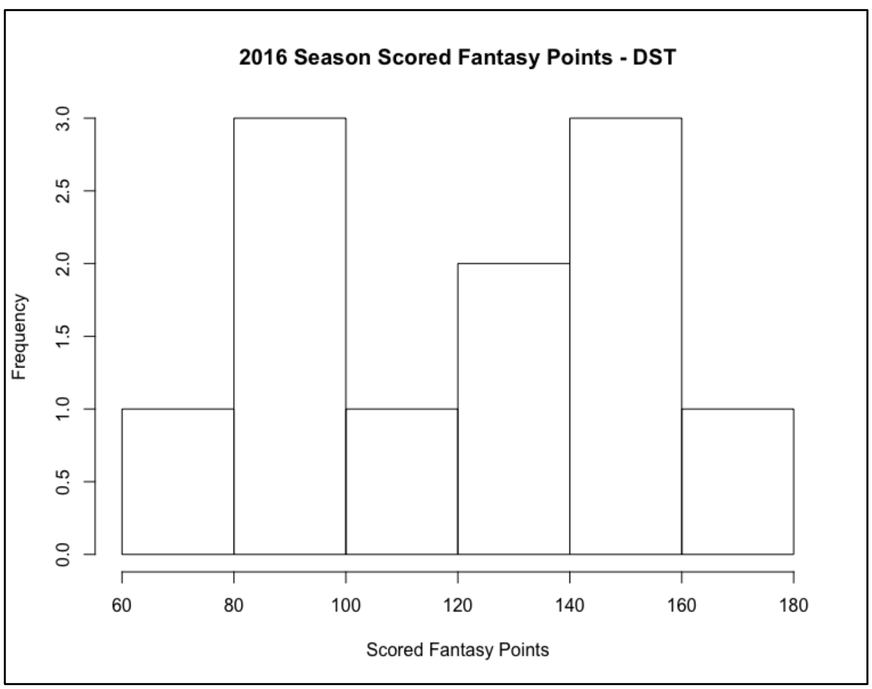

Figure 10. 2016 Season Scored Fantasy Points - D/ST

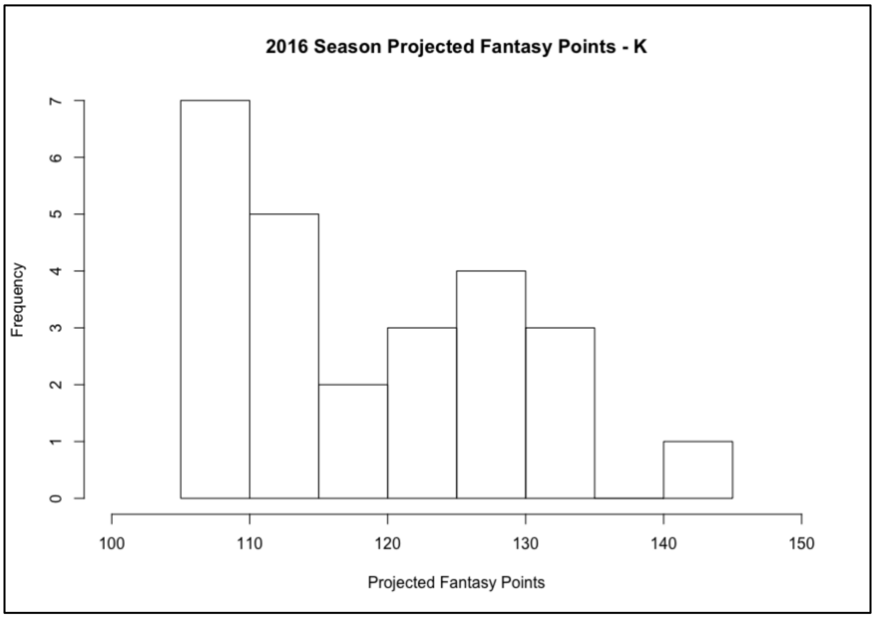

Figure 11. 2016 Season Projected Fantasy Points - K

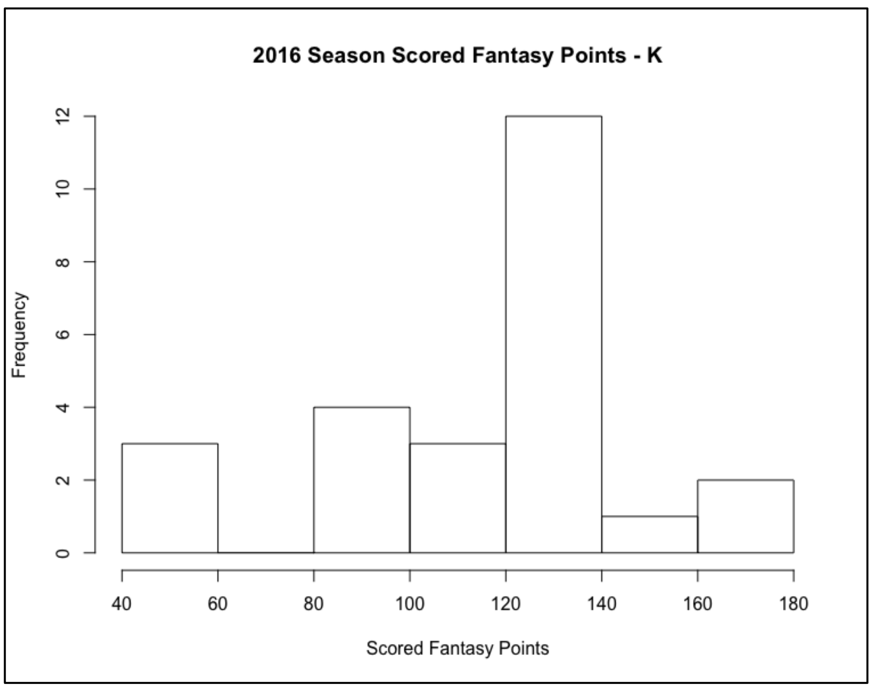

Figure 12. 2016 Season Scored Fantasy Points – K

Looking first at the quarterback position in Figure 1 and Figure 2, we can see that the season projections for 2016 were largely accurate. One player scored a very low number of points, and one player scored a very high number of points; in general, though, the top 18 quarterbacks performed about as well as projected over the course of the 2016 season. Summary statistics for both projected and actual quarterback scoring are given below in Table 4.

Table 4. 2016 Season Summary Statistics - QB

Points Type	***N***	***Min***	***Max***	***Range***	***Median***	***Mean***	***Standard Deviation***	***SE Mean***
Projected	18	241.7	346.1	104.3	279.1	284.3	27.4	6.5
Scored	18	224.2	380.0	155.8	264.35	274.8	36.0	8.5

Next looking at the running back position in Figure 3 and Figure 4, we can see that there were some relatively substantial deviations from projected points. Many players were projected to score between 50 and 80 points, and while many did score in this range, a substantially higher number than expected scored between 0 and 50 points. In addition, two players—Ezekiel Elliot and David Johnson—scored well above their projections and even well above the highest projections overall. Elliot was projected to score 205.3 points but scored 293.4, and Johnson was projected to score 209.0 but scored 327.8. For reference, the highest projected score of any running back going into 2016 was Todd Gurley, projected to score 222.3 points. Both Elliot and Johnson eclipsed this mark substantially. With the exceptions of players on the tail-ends of scoring, the projections were largely accurate. Summary statistics for both projected and actual running back scoring are given in Table 5.

Table 5. 2016 Season Summary Statistics - RB

Points Type	***N***	***Min***	***Max***	***Range***	***Median***	***Mean***	***Standard Deviation***	***SE Mean***
Projected	60	57.7	222.3	164.6	123.0	123.5	46.8	6.0
Scored	60	18.1	327.8	309.7	105.1	119.3	73.2	9.5

Now looking at wide receiver in Figure 5 and Figure 6, we see that the projected number of points for receivers was a right-skewed distribution, with many players projected to score between 50 and 100 points and only a few projected to score over 150. Figure 6 shows that the actual scored number of points follows something that much more resembles a normal distribution, although many players still do score between 50 and 100 points. Many players score between 100 and 150, mirroring the projections well. Even the small number of receivers who scored very highly mirrors the projected scores. While the distributions of the projected and scored fantasy points for wide receivers may not appear to be all that similar, a high concentration of receivers scoring in a very small point bandwidth creates this effect. In fact, though, the projections very accurately matched the scored points for receivers. Summary statistics for wide receivers' projected and actual scoring are given in Table 6.

Table 6. 2016 Season Summary Statistics - WR

Points Type	***N***	***Min***	***Max***	***Range***	***Median***	***Mean***	***Standard Deviation***	***SE Mean***
Projected	75	59.7	229.8	170.1	115.6	118.1	40.7	4.7
Scored	75	19.6	208.1	188.5	106.4	105.9	44.8	5.2

Now looking at tight ends in Figure 7 and Figure 8, we can see that the projected points scored appear to be an approximately normal distribution with a center just below 100 points. Looking at the scored points, though, we see a much more uniform distribution. No tight end scored quite as highly as the maximum projections, and no tight end scored quite as low as the minimum projections. Summary statistics for tight ends' projected and actual scoring are given in Table 7.

Table 7. 2016 Season Summary Statistics - TE

Points Type	***N***	***Min***	***Max***	***Range***	***Median***	***Mean***	***Standard Deviation***	***SE Mean***
Projected	25	29.2	173.9	144.7	89.7	96.7	30.4	6.1
Scored	25	20.5	138.0	117.5	79.2	80.6	34.4	6.9

Now looking at defense/special teams units in Figure 9 and Figure 10, there's admittedly little to be gained from comparing the projected points and actual scored points. The final scored points largely mirror the projected points, with a few points falling higher than the maximum projections and a few points falling lower than the minimum projections. Summary statistics for D/ST units' projected and actual scoring are given in Table 8.

Table 8. 2016 Season Summary Statistics - D/ST

Points Type	***N***	***Min***	***Max***	***Range***	***Median***	***Mean***	***Standard Deviation***	***SE Mean***
Projected	11	92.0	121.0	29.0	113.0	110.3	9.5	2.9
Scored	11	62.0	166.0	104.0	123.0	119.9	32.4	9.8

Finally, looking at kickers in Figure 11 and Figure 12, we see a right-skewed distribution for the projected points and a much more abnormal distribution for the actual number of points scored. Most kickers were projected to score between 110 and 135 points, with a few projected to be above this maximum and some projected to be slightly below the minimum. In reality, though, nearly all kickers really did score in a central band between 100 and 140. A few kickers scored very low, fewer points than 60, and a few kickers scored very high, above 160. In general, though, kickers seemed to score around the center of the projected scored for their position. Summary statistics for kickers' projected and actual scoring are given below in Table 9.

Table 9. 2016 Season Summary Statistics - K

Points Type	***N***	***Min***	***Max***	***Range***	***Median***	***Mean***	***Standard Deviation***	***SE Mean***
Projected	25	107.0	142.0	35.0	117.0	119.2	10.3	2.1
Scored	25	42.0	170.0	128.0	125.6	116.7	31.6	6.3

3.4.2 Week-to-Week Variance

One of the key aspects of building a good fantasy football team is the consistency of players. If two wide receivers score the same number of points per week on average but one is much more volatile than the other, the “safer asset”—the player with the lower variance—is the more desirable option for most teams.

Looking again at the top number of players, as dictated in Table 3, we can look at the week-to-week variance for players at each position during the 2016 NFL season. It’s important to note that players who played in every game of the season have 16 total data points for the season; any players who have fewer data points were either injured or removed from a game by their coach.

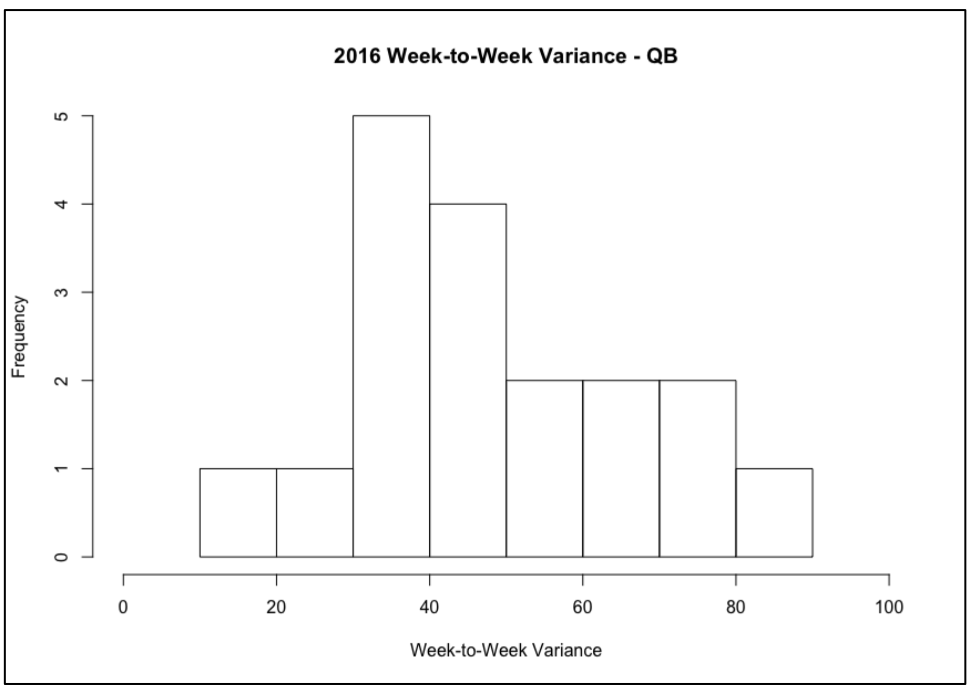

Figure 13. 2016 Week-to-Week Variance - QB

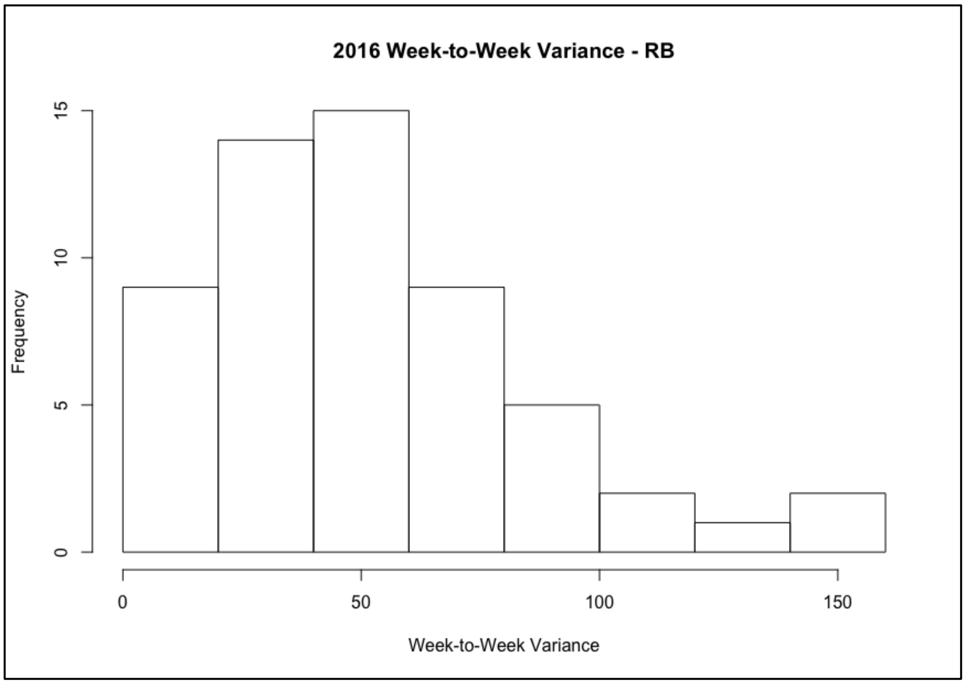

Figure 14. 2016 Week-to-Week Variance – RB

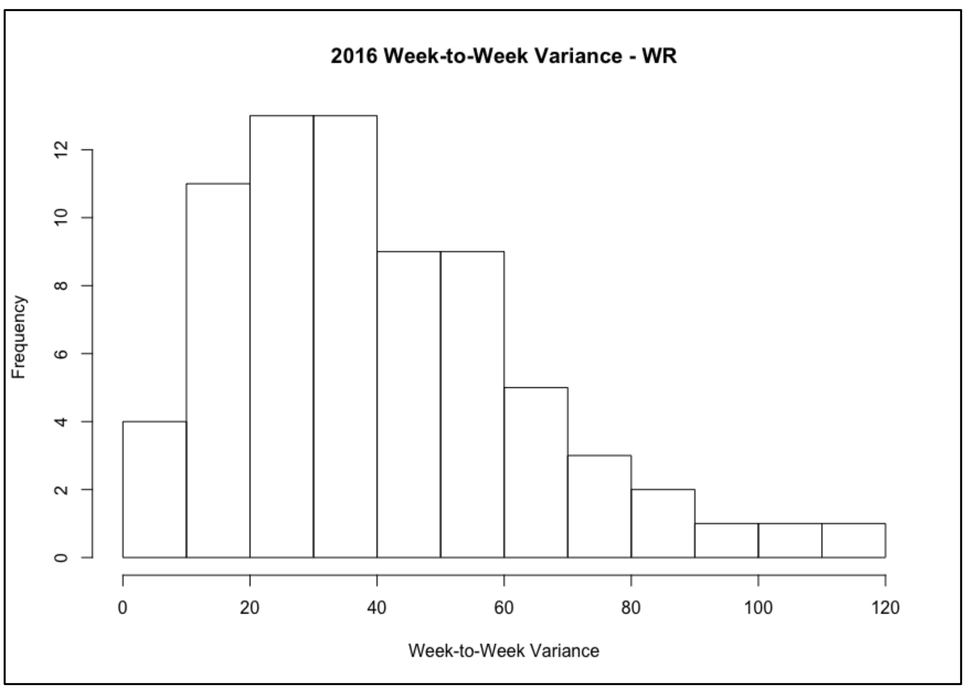

Figure 15. 2016 Week-to-Week Variance – WR

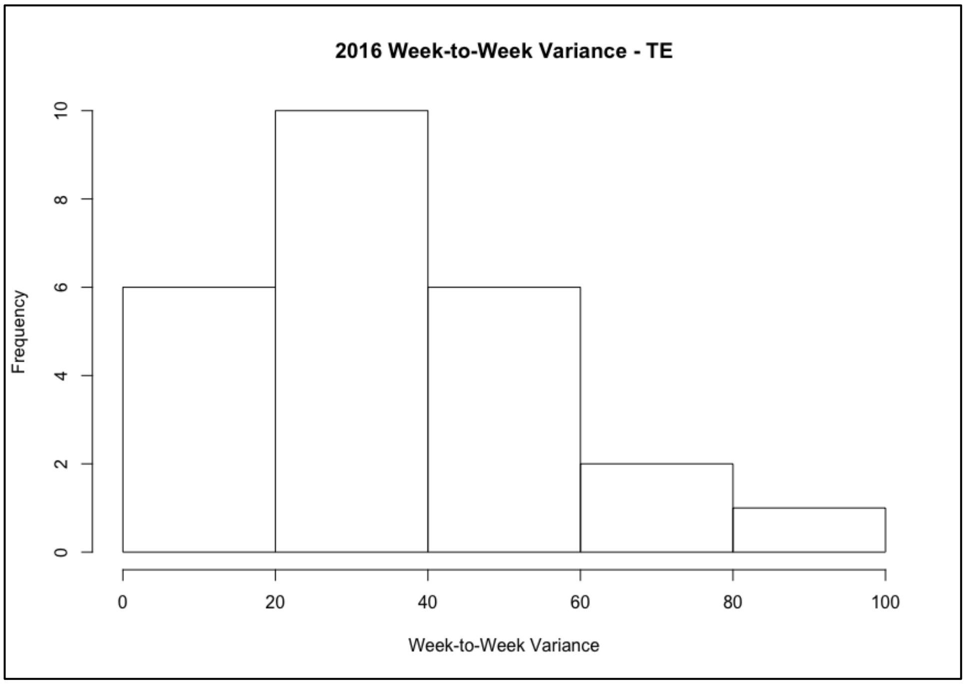

Figure 16. 2016 Week-to-Week Variance – TE

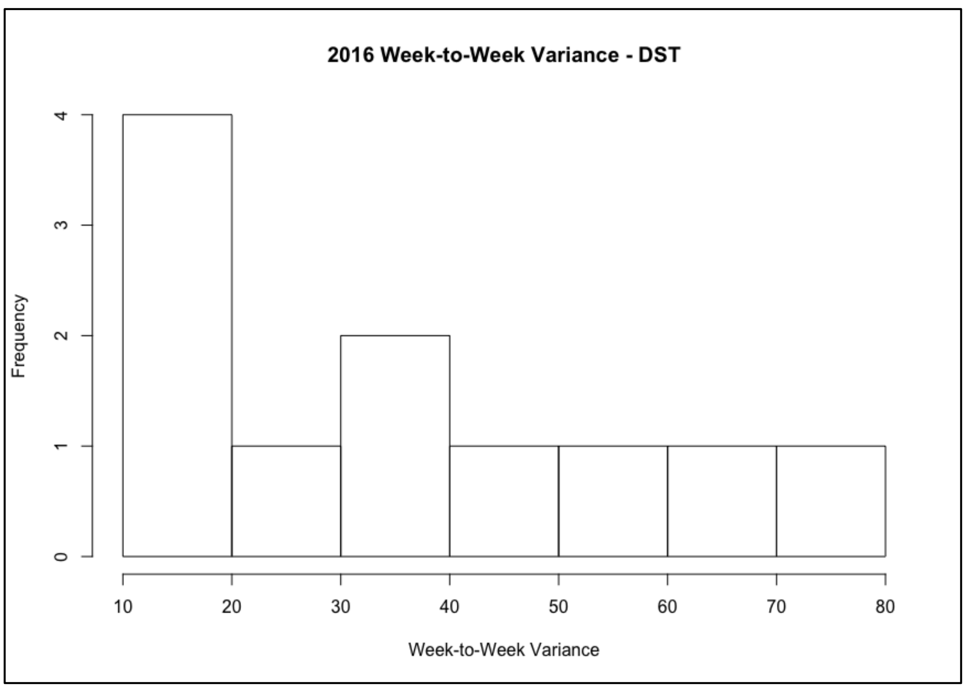

Figure 17. 2016 Week-to-Week Variance - D/ST

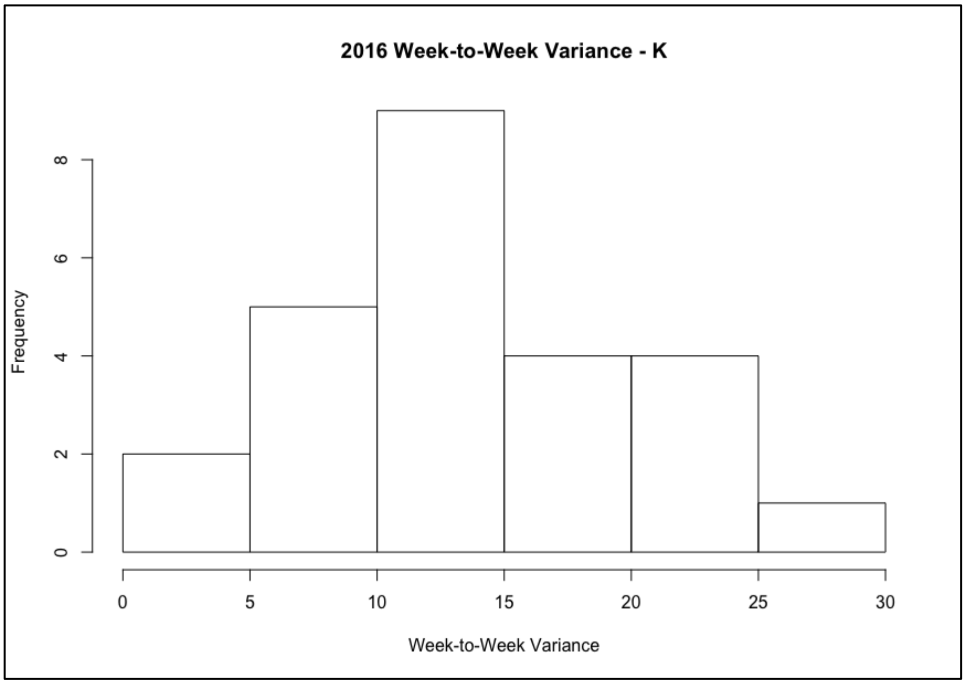

Figure 18. 2016 Week-to-Week Variance – K

Different positions show different levels of skewed-ness in the distributions of their week-to-week variance. Kickers appear to have a relatively normal week-to-week variance, while the distribution of week-to-week variance for wide receivers, for instance, appears to be right-skewed. Regardless of this difference between positions, Figures 13, 14, 15, 16, 17, and 18 each show that the variance does differ substantially among players of the same position. Since consistency is an important factor in building a fantasy football team, these differences can be exploited during the draft process in order to select an optimal team.

3.4.3 Mean-Variance Tradeoff

While looking at the consistency of a player's performance is essential for team-building, games are not won and lost by the variance of a team's performance; obviously, games are won and lost by how many points each team scores. Thus, the primary metric by which a player should be assessed is their average number of fantasy points scored.

Pairing the expected number of points scored with week-to-week variance should allow us to assess players more adequately. The primary metrics by which the draft optimization model (introduced in Chapter 3) selects players to be added to a fantasy team are the projected average score of a player and that same player's week-to-week variance. This mean-variance tradeoff is essential for selecting a fantasy team that scores highly and does so consistently.

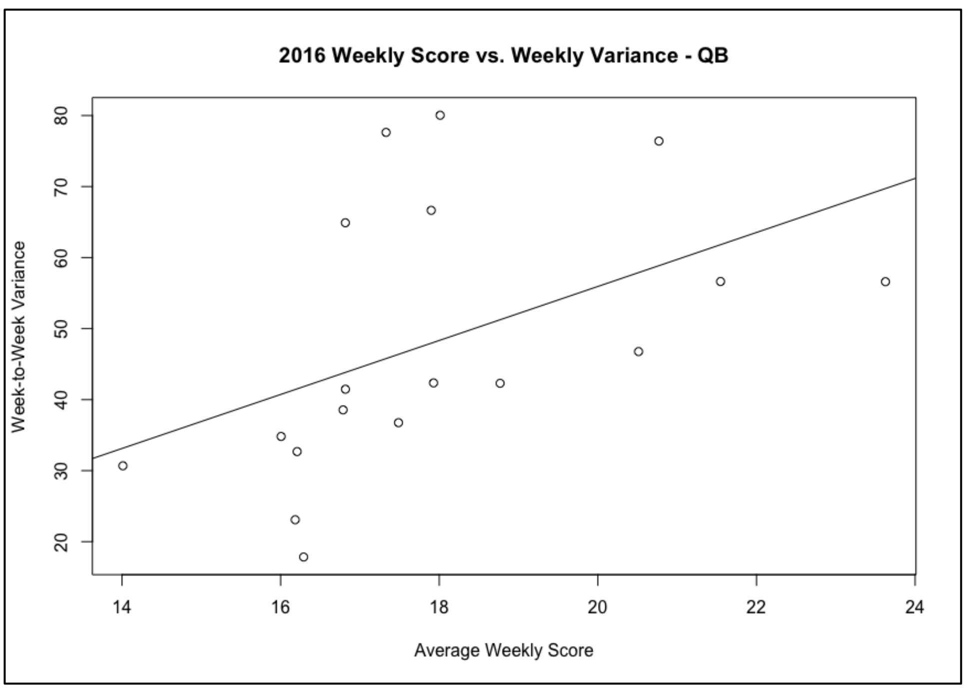

Figure 19. 2016 Average Weekly Score vs. Weekly Variance – QB

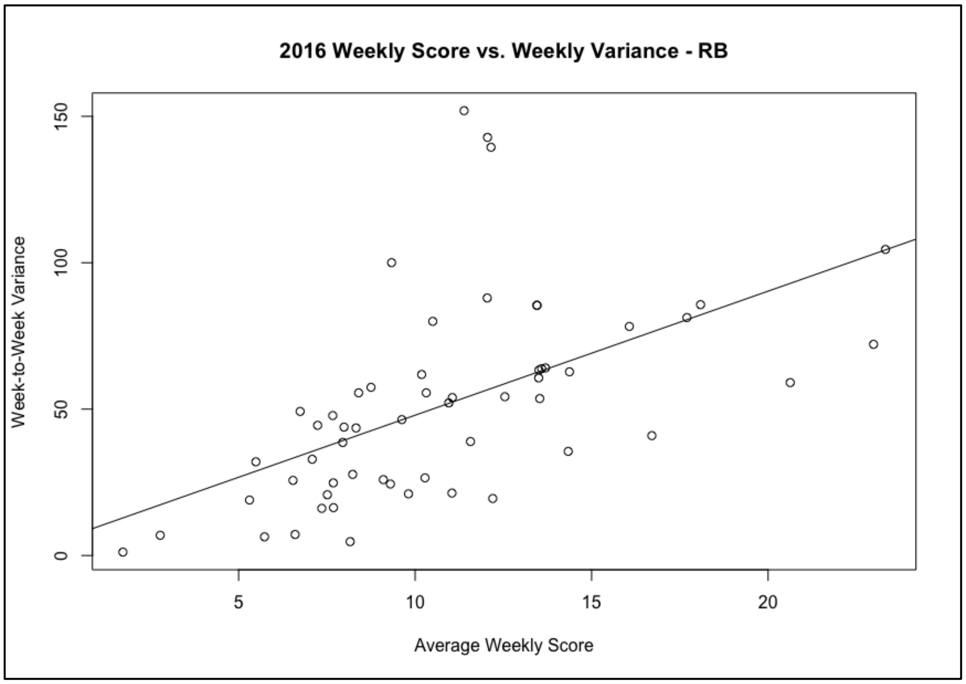

Figure 20. 2016 Average Weekly Score vs. Weekly Variance – RB

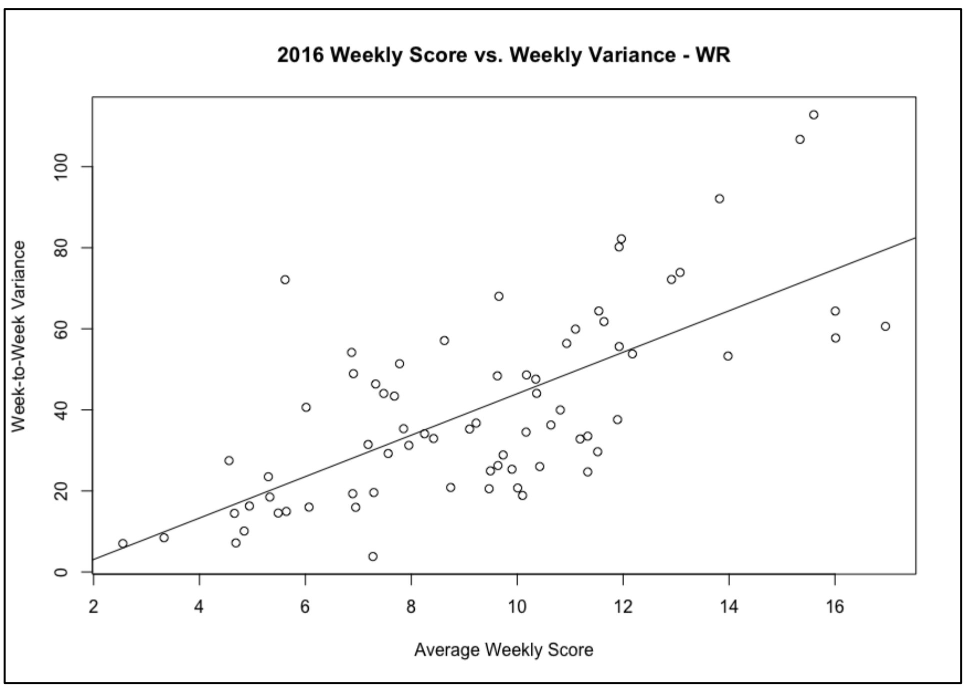

Figure 21. 2016 Average Weekly Score vs. Weekly Variance – WR

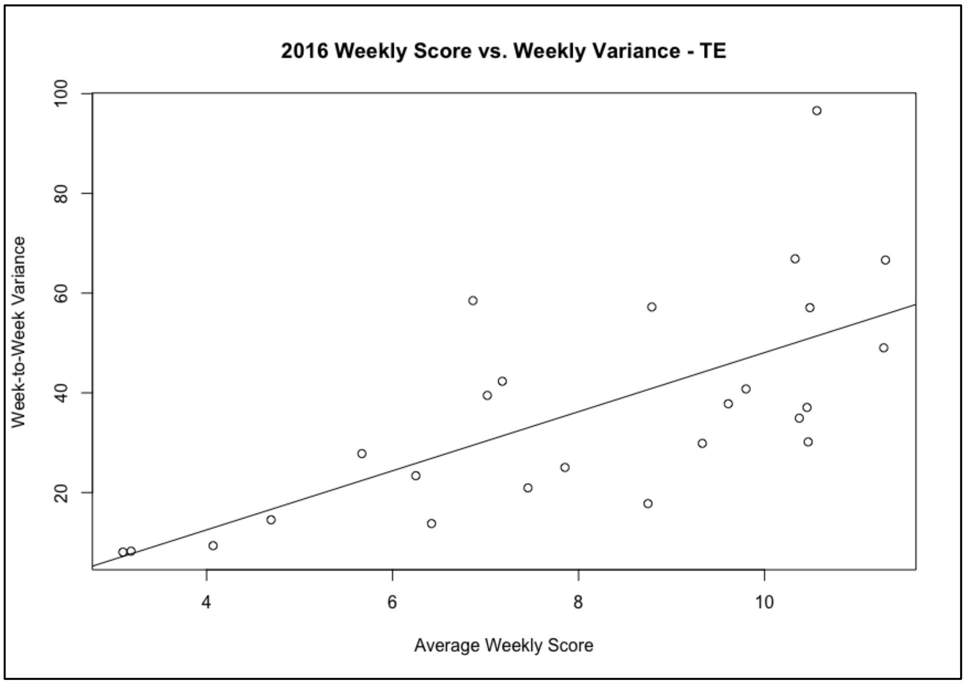

Figure 22. 2016 Average Weekly Score vs. Weekly Variance – TE

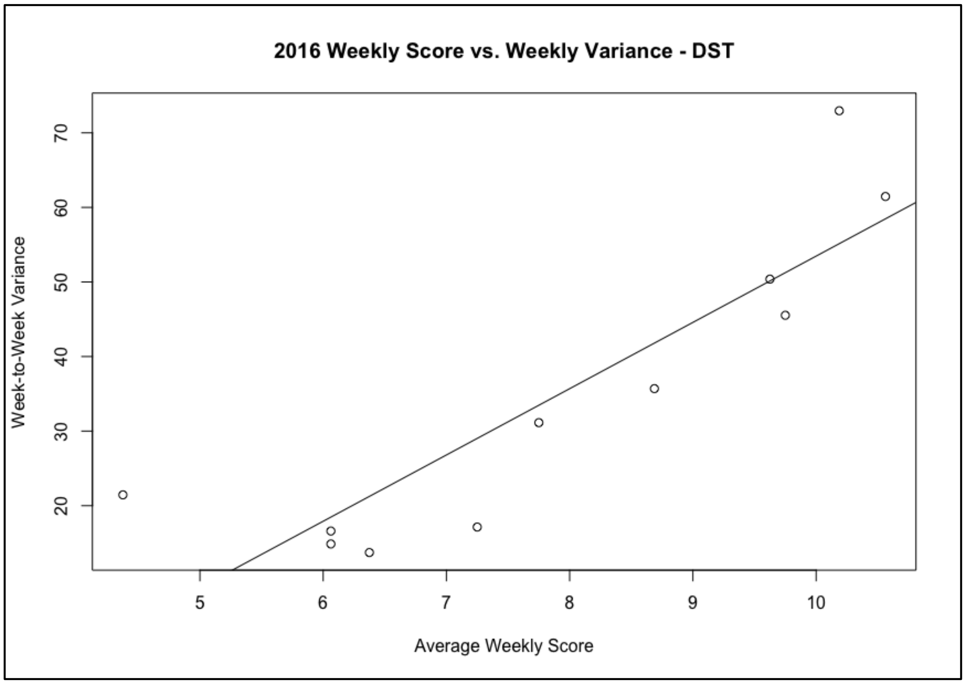

Figure 23. 2016 Average Weekly Score vs. Weekly Variance - D/ST

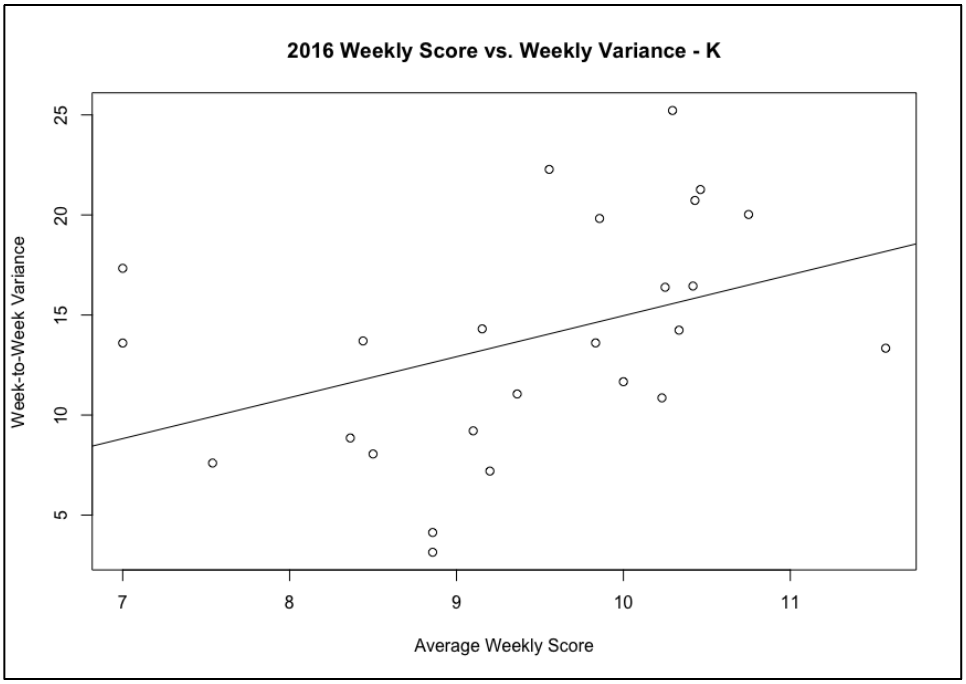

Figure 24. 2016 Average Weekly Score vs. Weekly Variance – K

Unsurprisingly, each position seems to show the same general trend pertaining to average weekly score and average week-to-week variance: as a player's average weekly score goes up, so does their average weekly variance. Looking at the best-fit trend lines added to each graph, this phenomenon is confirmed—players who score more points on average each week tend to have a higher variance.

However, not every player follows the trend for their position as much as other players. For each position, there are players who score around the average for their position but have a higher variance than would be expected by that average score; these players fall above the trend line in the graphs of weekly score vs. weekly variance. In contrast, there are also players who score a high number of points but who have a lower than expected variance based on the trend of their position; these players fall below the trend line in the graphs.

Players who score highly while having a low week-to-week variance are extremely valuable, while players who score very few points with a high week-to-week variance are much less valuable. Leveraging these statistics for each player should allow us to target players in the draft that allow the team to score highly and do so consistently.

3.4.4 Varying Numbers of Data Points for Each Player

One other important note in regards to the data used in this study relates to time. While some players have been in the NFL for over a decade, others only entered a few years ago, and others even have not yet played in an NFL game. Those who have played longer have somewhat of an advantage in our model. With more data points to consider, the calculated week-to-week variance should tend towards their actual variance, and their calculated weekly performance should tend towards their actual average output. Of course, a player with only one year in the league could be extremely consistent during that one year, but older players who have two or three times as many data points as younger ones may be advantaged in this model. This phenomenon is not necessarily a bad thing, as it results in players with longer track-records of success and consistency being prioritized, but it is still important to note the phenomenon. The weekly variances for players as well as average weekly scores will be calculated from data from the 2016, 2017, and 2018 NFL seasons. Any players who did not play in each of these seasons will simply have fewer data points from which to calculate those descriptive statistics.

Chapter 4

An Optimization Modeling Strategy for Draft Decision-Making

A bi-criteria optimization model will be developed in order to maximize the expected number of fantasy points scored and to minimize the week-to-week variance of player performance from players included on the team. In the long-term, a fantasy football team that scores a high number of points consistently should perform better than a team that has the potential to score a very high number of points but does so inconsistently.

This bi-criteria optimization modeling strategy will be used to assist the TO in making decisions for each of their individual draft picks. For the entirety of the draft, the TO will follow a draft selection algorithm which extends the effectiveness of the optimization model over the full length of the fantasy draft.

4.1 Bi-Criteria Modeling Background and Application

In the realm of financial engineering and portfolio optimization, a rational investor would most likely be interested in choosing a portfolio of investments that maximizes the returns of that portfolio but also minimizes the variance of that portfolio's return (Griffin, Prabhu, & Ravinadran, 2018). Mapping this premise to fantasy football, a TO should be interested in choosing a fantasy team—a portfolio of player investments—that maximizes the returns—points scored—of that team but also minimizes the variance of that team's performance.

In papers published in 1952, 1956, and 1959, Harry Markowitz explained his rationale for recognizing risk as an essential consideration in portfolio optimization, developed his own algorithm for solving general portfolio selection problems, and laid out the justification for basing portfolio optimization on a mean-variance analysis of available securities (Markowitz, 2000).

Markowitz proposed to maximize average annual return of the portfolio:

$$Z_1 = \sum_{j=1}^{N} \mu_j X_j = \mu^T X$$

Minimize the variance of the portfolio return:

$$Z_2 = \sum_{i=1}^{N} \sum_{j=1}^{N} {\sigma_{ij}}^2 X_i X_j = X^T Q X$$

Subject to the constraints:

$$\sum_{j=1}^{N} X_j \leq C; X \geq 0$$

$$\sum_{j \in J_1}^{N} X_j \leq b_1$$

$$\sum_{j \in J_2}^{N} X_j \geq b_2$$

where:

X_j = dollars invested in security j

X = $(X_1, X_2, \ldots, X_N)_T$ = investor portfolio

μ_j = average annual return per dollar in security j

$Q_{(NxN)} = [\sigma_{ij2}]$

N = total number of securities available for investment

C = total capital available for investment

This model set forth by Markowitz will be the building block upon which the draft optimization model and algorithm will be built. There are obvious issues with attempting to map the Markowitz model directly to one that satisfies the complexity of fantasy football player selection at the draft phase; accordingly, additional constraints will be added in order to account for the fantasy football league rules, logical relationships used in draft strategy, and other differences that may not be directly accounted for in the Markowitz model.

4.2 Parameters and Decision Variables of the Draft Optimization Model

Before addressing the entire draft selection algorithm or even the single draft choice optimization model itself, we'll provide the parameters and decision variables used in the model. It is important to note that players are indexed by *i* and positions are indexed by *j* in order to allow for more concise modeling. The number draft selection of the team owner is indexed by *k*.

4.2.1 Parameters

N: The set of draftable NFL players (i.e. those described by *M*) and defensive/special teams units.

M: The set of positions eligible to be drafted *M* = {QB, RB, WR, TE, K, D/ST}.

Pos(*i*): The position of player *i*, e.g. Pos(Saquon Barkley) = RB

WPosLimit(*j*): The maximum number of starting players for position $j \in M$ during any weekly contest, e.g. WPosLimit(K) = 1.

n_k: The overall pick number of the TO's *k*-th draft pick.

TOPlayer(*k*): The set of players that the TO has drafted after their *k*-th draft pick.

OppPlayer(k): The set of players that opponents have drafted after the TO's k-th pick.

$R_k(i)$: Out of the remaining draft-eligible players, the ranking of player i at the TO's k-th pick.

P(i): The projected number of points scored for the upcoming season by player i.

V(i): The previous week-to-week variance of player i.

SPosLimitU(j): The maximum number of players of position $j \in M$ which may be held on a fantasy team.

SPosLimitL(j): The minimum number of players of position $j \in M$ which may be held on a fantasy team; this also acts as the minimum number of players at position j that the TO must draft.

The variance parameter—V(i)—poses some issues in that player roles may change from year-to-year and alter the player's performance, rookies obviously have not yet played in any NFL games and thus have no week-to-week variance, and players who have been injured have exhibited a potentially low variance over a potentially significant portion of the sample size from which the variance would be calculated.

For the purposes of this model, player week-to-week variance is calculated based on player output from the 2016, 2017, and 2018 seasons. With 16 games per season, that should lead to a maximum of 48 data points to reference for each player. Even players who have missed a significant period of time due to injury should still have produced enough data points in order to calculate a week-to-week variance which is reflective of their typical performance level. As for rookies, because they have no NFL performances and thus no week-to-week variance, we assign them the average variance of their position for a first-year player from the previous season; in other words, we assign incoming rookies the average week-to-week variance of rookies from the previous year at the same position.

4.2.2 Decision Variable

The optimization model for the draft has one binary decision variable, which indicates whether the team owner has drafted the player or not.

$$y_i \in \{0,1\}: y_i = \begin{cases} 1, \text{if the TO selects player } i \\ 0, \text{otherwise} \end{cases}$$

4.3 The Draft Selection Optimization Model

The draft selection optimization model is a bi-criteria optimization model that seeks to maximize the total number of points scored over the course of the season by all players drafted by the TO while simultaneously minimizing the week-to-week variance of the players drafted. These objectives are subject to a set of constraints which describe fantasy football league rules, opponent draft behavior, and logical relationships that would affect draft decisions.

The result of the optimization model is a portfolio of players—including those already drafted and those projected to be the best available to draft in the future—which lies along an optimal frontier. In other words, considering the players already drafted by the TO, the draft optimization model yields a set of players which satisfies all roster constraints and would constitute an efficient portfolio.

For the fantasy team owner's k-th draft pick, the TO solves the following problem with objective functions given by Z_1 and Z_2:

$$\mathbf{max}\ Z_1 = \sum_{i \in N} P(i) y_i \tag{1a}$$

$$\mathbf{min}\ Z_2 = \sum_{i \in N} V(i) y_i \tag{1b}$$

Subject to the following constraints:

$$\sum_{i \in N : R_k(i) \leq \alpha \cdot (n_{\bar{k}} - n_k)} y_i \leq \bar{k} - k\ , \forall\ \bar{k} \geq k + 1 \tag{1c}$$

$$\sum_{i \in N : Pos(i) = j} y_i \geq SPosLimitL(j)\ , \forall\ j \in M \tag{1d}$$

$$\sum_{i \in N : Pos(i) = j} y_i \leq SPosLimitU(j)\ , \forall\ j \in M \tag{1e}$$

$$y_i = 1, \forall\ TOPlayer(k) \tag{1f}$$

$$y_i = 0, \forall\ OppPlayer(k) \tag{1g}$$

$$y_i \in \{0,1\}, \forall\ i \in N \tag{1h}$$

The objective functions (1a) and (1b) mirror the goals of the objective functions given in the previously discussed Markowitz bi-criteria optimization model. Equation (1a) states the objective of maximizing the total number of projected points scored in the upcoming season by all players drafted by the team owner. Equation (1b) states the objective of minimizing the total week-to-week variance of all players drafted by the team owner. This objective of simultaneous maximization of expected return and minimization of risk is at the core of the draft optimization model.

Constraint (1c) is used to help model the drafting strategy of opponent owners. Essentially, the constraint states that, up to the $\bar{k}$-th pick, the team owner can draft a maximum of $\bar{k} - k$ players from the top $\alpha \cdot (n_{\bar{k}} - n_k)$ players, as determined by the rankings in R_k. The pre-determined value of the constant α mimics the real-world uncertainty around the drafting decisions of opponent owners. With more unpredictable owners, the α may take on a higher value, while with more predictable owners who likely draft based on players rankings, the α value would likely take on a value closer to 1. The parameter $\alpha \geq 1$, along with the inclusion of player rankings in the draft algorithm, are the main methods used to mirror opponent draft strategy.

Constraints (1d) and (1e) describe roster constraints characteristic to the rules of the fantasy football league. Constraint (1d) enforces that the team owner must draft at least the minimum number of players at position *j* required by league rules; this lower bound is given by SPosLimitL(*j*). Similarly, constraint (1e) enforces that the team owner may draft at most the maximum number of players at position *j* allowed by league rules; this upper bound for position *j* is given by SPosLimitU(*j*).

Constraints (1f) and (1g) enforce the logical constraints that players already drafted by the team owner remain on the team, and players already drafted by opposing owners may not be selected to be drafted by the team owner.

Finally, constraint (1h) enforces the binary nature of the decision variable, y_i.

4.4 The Draft Selection Algorithm

The draft selection optimization model solves for team owner's optimal fantasy team at the TO's *k*-th pick given a set of players already drafted to the team and a set of players already drafted by opponents. Even with constraints in that optimization model aimed to mirror opponent draft strategy, there is no way to entirely predict the behavior of other fantasy team owners. In order to make the best decisions possible at every pick, the TO should solve the draft optimization model at each pick *k*, analyze the players indicated to be a part of the efficient team, and use the available player rankings to compare the model's suggestion to likely opponent draft choices.

4.4.1 The Draft Selection Algorithm

1. For each team owner draft pick k = 1, 2, …, n_D:

    2. Solve the draft optimization model. Denote the solution as y_{ik} for all $i \in N$.

    3. Draft the player not already on the team with the highest rank: $i_k = \arg\max_{i \in P_k} R_k(i)$.

    4. Update TOPlayer(k+1) = TOPlayer(K) $\cup$ {i_k}.

Essentially, this draft selection algorithm drafts the player not already on the team but included in the newly calculated optimal team selection with the highest player rank. The player of the highest rank is chosen as another mechanism of modeling and outsmarting opponent behavior. In choosing the player of the highest rank, the team owner ensures that they are drafting the player most likely to be taken by another fantasy team owner before their next draft pick. If this ranking method was not employed, the team owner could end up drafting a player who would have been available to draft at their next selection opportunity. Thus, the use of player rankings as a method of selection is another layer of draft strategy which ensures that the team owner chooses the best fantasy team in comparison to the teams of other fantasy owners.

4.5 A Numerical Application of the Draft Optimization Algorithm

As the draft optimization algorithm employs a bi-criteria linear program, the output will be a set of choices which form an efficient frontier. By finding end values for the objective functions and then using one objective as an additional constraint, we can form the efficient frontier. A simulated fantasy football draft will be used to produce teams along the efficient frontier, and the players on those teams will be compared through the lens of the objective functions.

4.5.1 The Effect of Draft Choice on Final Results

For this example, a 10-team league employing a “snake draft” drafting method is used. In order to isolate the effects of the draft optimization model, the effect of initial draft position is investigated first.

Simulating a draft in which the fantasy team owner begins with the first pick yields a total projection of 2,584 points and an accompanying team variance of 887. Simulating other drafts in which the team owner begins with picks two through ten yields similar results. In fact, the initial draft position with the lowest projected total points is tenth, which yields a projection of 2,564 and variance of 802. The projected points scored by the team and the total team variance for a team owner seeking to maximize points scored are shown below in Table 10.

Table 10. Projected Points and Team Variance by Initial Draft Pick

Initial Draft Pick	***Total Projected Points***	***Total Team Variance***
1	2,584	887
2	2,582	897
3	2,575	832
4	2,581	879
5	2,575	830
6	2,581	826
7	2,583	862
8	2,576	814
9	2,569	809
10	2,564	802

4.5.2 Defining the Efficient Frontier of Fantasy Team Portfolios

The efficient frontier of fantasy portfolios is a set of pairings of team projected scoring and team variance which simultaneously maximize scoring—defined by Z_1—and minimize total variance—defined by Z_2. In order to define the efficient frontier, Z_1 was first maximized subject to all constraints included in the model; however, Z_2 was not considered, allowing the total variance to grow without bound. This true maximization of total projected points yielded a result of 2,584 expected fantasy points and a total team variance of 887. Next, Z_2 was minimized subject to all constraints in the model; however, Z_1 was not considered, allowing the projected number of points scored to shrink without bound. This true minimization of total team variance yielded a result of 730 expected fantasy points and a total team variance of 91. The two mean-variance pairs found by maximizing total projected points and then minimizing total team variance served as endpoints for further study.

Using increments of 50 for the variance, an upper bound was placed on the total team variance—beginning at 100 and moving up to 900. In other words, in practice, an additional constraint was added to the model, with *l* representing the incrementally increased upper bound.

$$Z_2 = \sum_{i \in N} V(i) y_i \leq l$$

With the constrained variance in place, the optimal teams which produced the greatest number of projected fantasy points were found. Because these teams essentially minimized the total team variance as well, they sat along the efficient frontier for fantasy football teams. Using a variance step size of 50, 18 total mean-variance pairs were found using this technique. An open source Excel add-on called Open Solver (Mason 2012) allowed for these simulations with a total of 324 binary decision variables—y_i. Each mean-variance pair represented a unique set of 16 NFL players, and each of these sets satisfied the full list of constraints outlined in Section 4.3. Table 11 below shows the team variance and total number of projected fantasy points from simulations at each variance step.

Table 11. Projected Score and Total Variance of Efficient Portfolio Teams

Maximum Team Variance	***Total Team Variance***	***Total Projected Fantasy Points***	***Maximum Team Variance***	***Total Team Variance***	***Total Projected Fantasy Points***
NA	91	730	500	498	2450
100	99	847	550	550	2493
150	147	1381	600	597	2519
200	199	1678	650	645	2544
250	248	1889	700	700	2562
300	300	2072	750	746	2572
350	350	2223	800	799	2579
400	398	2329	850	843	2583
450	449	2400	NA	887	2584

4.5.3 Measuring the Tradeoff Between Projected Score and Variance

Unsurprisingly, the bi-criteria modeling technique did not yield a dominant point which maximized projected total points scored while simultaneously producing the minimum value for total team variance. As a result, building the efficient frontier using the data from Table 11 was the appropriate next step in measuring the tradeoff between projected score and team variance. Graphing the efficient frontier showed the connection between maximizing projected points scored and minimizing total team variance, as well as the tradeoff between the two. Figure 25 below shows this tradeoff.

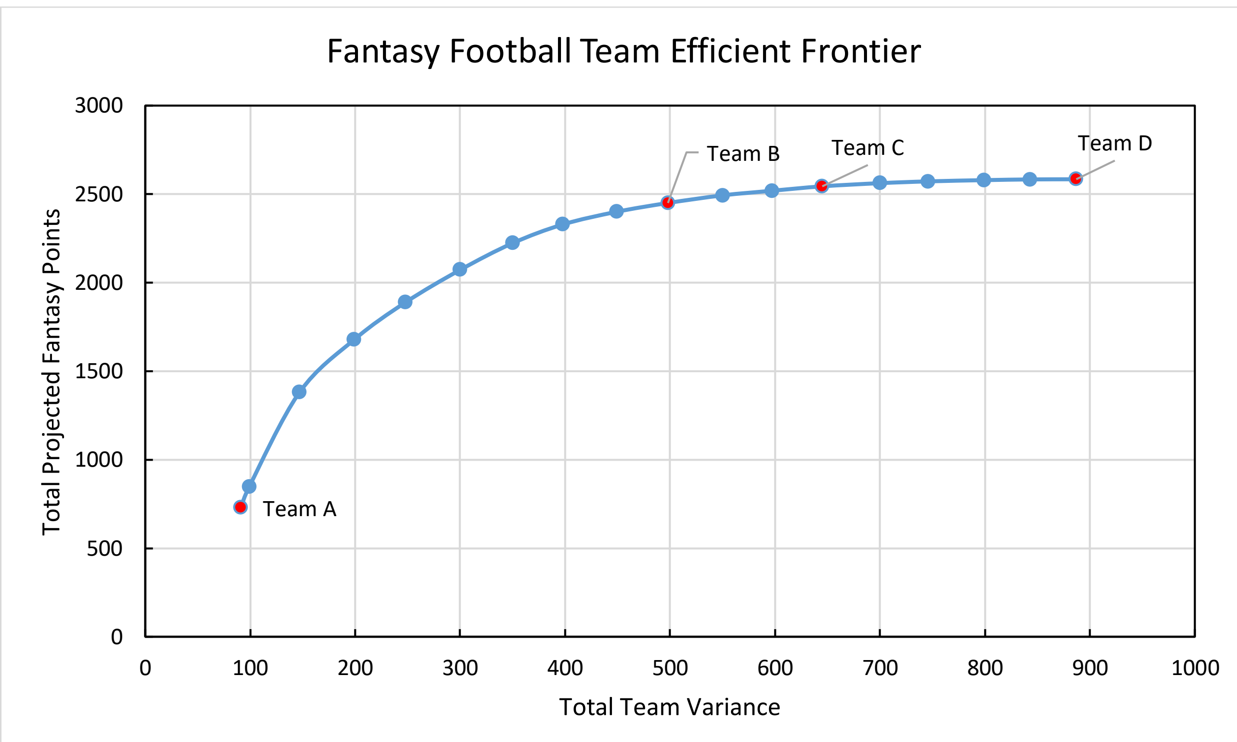

Figure 25. Efficient Frontier of Fantasy Football Teams

As we can see, for teams with extremely low variances, total projected points scored also tend to be low. As the allowed variance increases through about 400 or 500, the total projected points increase substantially. From that point on, though, the curve seems to level off, suggesting that there may be a point of diminishing returns of projected points as we increase the allowed total team variance. There is clearly a substantial tradeoff between team variance and the total projected scoring, but choosing a team along the efficient frontier still comes down to the risk preferences of the team owner. A more “aggressive” owner would likely favor the teams with higher projected points scored, even though they come with higher variances. A more “conservative” owner would likely favor the teams with a slightly lower projected score, as they seem to come with substantially lower total variance figures.

4.5.4 A Comparison of Fantasy Teams Along the Efficient Frontier

Four specific points along the efficient frontier were chosen for further analysis; these points are identified on Figure 25 by red points and are marked as Team A, Team B, Team C, and Team D.

Team A represents the absolute minimization of total team variance without regard for the maximization of total projected scoring. Team B represents the mid-point of the variance steps taken in creating the efficient frontier, falling halfway between the minimum total team variance defined by Team A and the maximum total team variance defined by Team D. Team C was chosen as a point of comparison, as it seemed to be a point which very nearly maximized the total number of projected points scored but yielded a substantially lower variance than the truly maximized total points projection; in addition, Team C appeared to be a point along the efficient frontier before the curve began to dive off steeply towards lower returns with lower variances. Finally, Team D represents the absolute maximization of total team points without regard for minimizing total team variance. The projected points, total team variance, and team rosters for Team A, Team B, Team C, and Team D can be found in Table 12 below.

Table 12. Comparison of Fantasy Teams Along the Efficient Frontier

Team A	***Team B***	***Team C***	***Team D***
784 projected fantasy points	**2450 projected fantasy points**	**2544 projected fantasy points**	**2584 projected fantasy points**
91 total team variance	**498 total team variance**	**645 total team variance**	**887 total team variance**
James Washington (WR)	Christian McCaffrey (RB)	Christian McCaffrey (RB)	Christian McCaffrey (RB)
Cleveland (DST)	Chris Carson (RB)	Leonard Fournette (RB)	Leonard Fournette (RB)
Dare Ogunbowale (RB)	Kerryon Johnson (RB)	Chris Carson (RB)	Derrick Henry (RB)
Darren Waller (TE)	Tyler Lockett (WR)	Marlon Mack (RB)	Marlon Mack (RB)
Malcolm Brown (RB)	Julian Edelman (WR)	Tyler Lockett (WR)	Tyler Lockett (WR)
Justin Watson (WR)	Baker Mayfield (QB)	Baker Mayfield (QB)	Calvin Ridley (WR)
Chris Moore (WR)	Carson Wentz (QB)	Alshon Jeffery (WR)	Allen Robinson (WR)
Chad Beebe (WR)	Curtis Samuel (WR)	Kenyan Drake (RB)	Kenyan Drake (RB)
Keith Kirkwood (WR)	Rashaad Penny (RB)	Rashaad Penny (RB)	Sammy Watkins (WR)
Lee Smith (TE)	Sterling Shepard (WR)	Cam Newton (QB)	Cam Newton (QB)
Ryan Switzer (WR)	Royce Freeman (RB)	Sterling Shepard (WR)	Marvin Jones (WR)
Mike Gesicki (TE)	Peyton Barber (RB)	Peyton Barber (RB)	Dante Pettis (WR)
Daniel Carlson (K)	Courtland Sutton (WR)	Courtland Sutton (WR)	Dak Prescott (QB)
Taysom Hill (QB)	New England (DST)	New England (DST)	New England (DST)
Dontrell Hilliard (RB)	Stephen Gostkowski (K)	Stephen Gostkowski (K)	Stephen Gostkowski (K)
Chris Lacy (WR)	Darren Waller (TE)	Kyle Rudolph (TE)	Greg Olsen (TE)

Looking first at Team A, we see an extremely low projected points scored to go along with the absolute minimum total team variance. In truth, this team would certainly not be a viable option to draft, as the players on it simply do not yield enough fantasy points to win a fantasy football league.

Looking next at Team B, we can see a relatively high number of projected fantasy points—2,450—to go along with a variance of only 498. The top three players on this team are Christian McCaffrey, Chris Carson, and Kerryon Johnson. McCaffrey ranks 3rd, Carson ranks 23rd, and Johnson ranks 30th, according to ESPN's fantasy player rankings. McCaffrey had a projected season point total of 247.9 and accompanying variance of 78.75, Carson had a projected score of 189.2 with a variance of 42.26, and Johnson had a projected score of 177.4 and impressively low variance of only 29.41. Looking at these three players, we can certainly see the tradeoff between points scored and total variance. Players with substantially higher score projections are included in the team portfolio because they so significantly improve the total team projection, while other roster spots are taken by players with slightly lesser projections but substantially better consistency.

Looking next at Team C, we can see a high number of projected total fantasy points—2,544—to go along with a variance of 645. This team would absolutely be viable in a fantasy football league. Top players Christian McCaffrey, Leonard Fournette, and Chris Carson rank 3rd, 21st, and 23rd respectively. McCaffrey had a projected season point total of 247.9 and accompanying variance of 78.75, Fournette had a projected total of 191.1 points with a variance of 56.92, and Carson had a projected score of 189.2 with a variance of 42.26. Looking just at these three players, we can see the tradeoff between projected points scored and demonstrated variance. McCaffrey is a rare commodity and projects to score significantly higher than other players, but he also shows a high variance. Fournette and Carson, on the other hand, show high score projections but substantially lower total variances.

Looking finally at Team D, we can see the highest number of possible projected fantasy points—2,584—to go along with a variance of 887. This team would be the highest projected scoring one in a typical fantasy league, which is obviously an intriguing prospect for a team owner. However, because of its high variance, the team would be much more likely to score a very high number of points one week and a low total the following week; in other words, the team may lose weekly contests because it performs so poorly in certain weeks, although it may dominate other contests. The top three players on Team D are Christian McCaffrey, Leonard Fournette, and Marlon Mack. McCaffrey has a projected point total of 247.9 and variance of 78.75, Fournette has a projected total of 191.1 and variance of 56.92, and Marlon Mack has a projected point total of 175.1 and variance of 61.06. Entirely unconstrained by variance, this team has a large number of players who project to score extremely high point totals, although their inconsistent performances also lead to high week-to-week variances. For a team owner simply looking to score the highest number of points over the course of a full season, Team D may be the best option, although for a team owner looking for consistency, Team D certainly does not provide that.

Looking at the rosters of each team, we can actually see that there are a number of players shared between the teams. This isn’t too surprising, since each team had the same constraints and same general objectives of maximizing total score and minimizing total variance. Table 13 below shows the number of players shared between each pair of teams.

Table 13. Number of Players Shared by Efficient Teams

Team Pairing	***Number of Shared Players***
Team A and Team B	1
Team B and Team C	10
Team C and Team D	8
Team A and Team D	0
Team A and Team C	0
Team B and Team D	4

Team A and Team B share only one player, and Team B is actually the only team with which Team A shares a player. This is unsurprising, since Team A is far removed from the other teams in terms of the projected number of points scored and even the total variance.

Team B and Team C share 10 of 16 players, which is also unsurprising given the fact that their only difference in modeling was a maximum variance of 500 for Team B and 650 for Team C. Team C does project to score 94 points more over the course of the season than Team B, the equivalent of 5.88 points per week in a typical 16-week season. This difference is certainly not negligible, but in order to curtail Team C's variance from 650 to 500, only six players from a roster of 16 needed to be changed.

Team C and Team D share 8 players out of the 16 on each of their rosters. This comparison is by far the most telling, as Team D produced the highest projected points but also the highest total team variance. Team C, on the other hand, cut the team variance from 887 to 645 while only losing a few projected points scored—moving from 2,584 to 2,544. Over the course of a 16-week season, this 40 total point difference equates to an average difference of 2.5 points per week. Team

C projected to score 159 points per weekly contest, falling only 2.5 points short of Team D's performance; however, Team C also carries a substantially lower variance than Team D. Team C's projected points scored is only 1.5% lower than Team D's, but Team C's total variance is 27.3% lower than Team D's.

4.5.5 Conclusions

The mean-variance tradeoff illustrated specifically by Team C and Team D is exactly what this research sought to demonstrate. By taking on only a negligible 1.5% decrease in the projected number of points scored, Team C also improves on Team D's consistently, lowering the total team variance by a substantial 27.3%. In terms of fantasy league play, there is certainly more value in having a consistently high scoring team such as Team B or Team C, as opposed to a "boom or bust" team like Team D.

Additionally, the fact that Team B and Team C share 10 players and Team C and Team D share 8 shows that improving a team by minimizing its variance is not all that difficult. Again comparing Team C and Team D, the difference between the two is that Team C, at the same decision points as Team D, chose more consistent players in the draft. By using this bi-criteria optimization technique, Team C identified a comparably scoring roster to that of Team D with a much more consistent output. This combination of high scoring and consistent output is exactly the finding that this research set out to find.

Chapter 5

An Optimization Strategy for In-Season Roster Management

Weekly roster management entails two primary tasks. First, the team owner must decide whether they should add a player from free agency (i.e. a player not owned by any other fantasy team owner) in exchange for dropping a player currently held on their team. Second, the team owner must decide which players from their roster should be included in their starting lineup for that week (i.e. which players from their roster will accrue points). In a season-long context, these two actions are undoubtedly inter-related. Dropping a player in week 3 precludes the team owner from including that player in their starting lineup in week 4. At the same time, there are certainly situations in which dropping players to free agency in exchange for an addition of another player is wise (e.g. the player to be dropped is injured and will not accrue points for the rest of the season). Both essential aspects of weekly roster management are discussed in this chapter.

5.1 In-Season Roster Optimization Strategy

Because the ideas of setting starting lineups from existing players on the roster is so intimately tied to the idea of adding and dropping free agents, one optimization model should be created to include both of these aspects. In other words, one optimization model should yield the optimal roster decision each week given the players on the roster, players available in the free agent pool, and the traits of each player.

The starting lineup optimization model is a linear program which seeks to maximize the total number of points scored by players in starting lineups over the course of the entire season. Considering all players on the roster, their projections, and their bye weeks, this model produces the expected maximum number of points scored by players on the roster when they are placed optimally into the starting lineup each week.

The free agency optimization algorithm uses the starting lineup optimization model to determine the "value added" to a team by adding a player from the free agent pool and subsequently dropping a player from the TO's roster. This algorithm determines the projected season-long impact of executing an "add/drop" move in a specific week, a long-term view which is often lost in making these types of roster decisions.

5.2 Parameters and Decision Variables of the Starting Lineup Optimization Model

The core aspect of in-season roster management is choosing which players from the roster the TO should have in the starting lineup each week. With seven bench players on a typical roster, there are undoubtedly difficult decisions relating to which players should be added to starting lineup and which should be left on the bench. The season-long starting lineup optimization model projects the highest-scoring starting lineup for each week. The output of the model is not only the projected optimal starting lineup policy for each week during the season, but it is also the projected point total for the entire season of those starting lineups.

5.2.1 Model Parameters

y_{ti}: A binary variable indicating whether player *i* is on the team owner's roster during week *t*.

x_{ti}: The projected score of player *i* during week *t*. In week *t*, this figure is based on published projections from fantasy host websites; otherwise, it is the average projected weekly score based on season projections (i.e. P(*i*)/T).

f_{ti}: A binary variable indicating whether player *i* acts as a "flex" player for the fantasy team's starting lineup in week *t*.

WPosLimitL(j): The minimum number of players at position *j* that may be included in a starting lineup during any week.

WPosLimitU(j): The maximum number of players at position *j* that may be included in a starting lineup during any week.

5.2.2 Decision Variable

a_{ti}: A binary decision variable indicating whether player *i* is included in the starting lineup of the team owner's fantasy team during week *t*.

5.3 The Starting Lineup Optimization Model

This optimization model uses simple greedy logic to choose the players from a team's existing roster who will score the most points as a starting lineup during each week; subsequently, the model also sums the optimized starting lineups from week *t* through week *T*. The result of the model is not only a projected number of points scored by the team's starting lineups through the rest of the season, but it is also a projected optimal policy for setting starting lineups based on the currently available roster.

$$\mathbf{max}\, Z_3 = \sum_{s=t}^{T} \sum_{i} y_s{}^i \cdot x_s{}^i \cdot a_s{}^i$$

(2a)

Subject to the following constraints:

$$\sum_{i \in TOPlayer_t : Pos(i) = j} y_t{}^i \cdot a_t{}^i \geq WPosLimitL(j), \forall\, j \in M$$

(2b)

$$\sum_{i \in TOPlayer_t : Pos(i) = j} y_t{}^i \cdot a_t{}^i \leq WPosLimitU(j), \forall\, j \in M$$

(2c)

$$1 = \sum_{i \in TOPlayer_t : Pos(i) = j} y_t{}^i \cdot a_t{}^i \cdot f_t{}^i, \forall\, j \in M = \{RB, WR, TE\}$$

(2d)

$$y_t{}^i, a_t{}^i, f_t{}^i \in \{0,1\}\ \forall\, i \in N, \forall\, t \in T$$

(2e)

The objective function given by (2a) gives the summation of fantasy points scored by all players included in the starting lineup over the course of the entire season from week *t* through week *T*. The inner summation finds the total score of the optimal starting lineup for a given week, and the outer summation sums the optimal scores from each week.

Constraints (2b) and (2c) impose roster-building logic on the model. Constraint (2b) states that the starting lineup must include at least the minimum number of players required for each position; constraint (2c) states that the starting lineup must include no more than the maximum number of players allowed at each position.

Constraint (2d) implements logic to describe the nature of the "flex" spot in the starting lineup. Each starting lineup includes one "flex" player, whose position may be either RB, WR, or TE. This player is added to the starting lineup in addition to the players already listed at the RB, WR, and TE positions. Constraint (2d) dictates that only one player—specifically a running back, wide receiver, or tight end—may be included in the starting lineup as a "flex" option.

Finally, constraint (2e) imposes the binary nature of terms y_{ti}, a_{ti}, and f_{ti}.

5.4 The Free Agency Optimization Algorithm

Fantasy football leagues cannot be won by solely making good decisions during the initial draft. Surely, these decisions can set a team up for success, but with factors such as injuries, player role changes, player trades within the NFL, players who "break out," and players who "bust," no fantasy team owner can expect to win a fantasy league solely based on decisions made at the draft or even decisions in setting starting lineups. Dropping players from their fantasy team who have little chance of contributing to the team in the future in favor of adding players from the free agent pool who may contribute is a necessary team management practice which substantially improves the chances of winning a fantasy football league.

With that said, the long-term ramifications of these "add/drop" decisions are often not understood by team owners. Perhaps adding a player from free agency may be advantageous for one week, but if the player dropped would have contributed in three weeks in the future, that roster move was likely not a wise one. In order to capture the season-long nature of these decisions, a sequential decision analysis model is created to maximize the total number of pointed scored by players in the starting lineup each week over the course of the entire football season.

We can view the season as a series of decision epochs at which the team owner must decide as to whether they will add a player from the free agency pool at the expense of dropping a player already on their fantasy team. The simplest way of determining a player's "value added" to the fantasy football team is by looking at the number of times that player projects to be in the team's starting lineup (i.e. the number of times that player should accrue points), summing that total number of points, and considering it against the similarly calculated contributions of other players over the course of the season.

5.5 Parameters and Decision Variables of the Free Agency Optimization Algorithm

The free agency optimization algorithm determines the "value added" by a player in the current free agent pool in comparison to a player on the TO's current roster. The algorithm uses the starting lineup optimization model to calculate the projected total number of points scored by starting lineups over the course of the season (i.e. through week *T*) when an "add/drop" move is executed in week *t*.

5.5.1 Model Parameters

$TOPlayer_t$: The set of players on the team owner's roster in week *t*.

$OppPlayer_t$: The set of players on the rosters of all opponents in week *t*.

$FAPlayer_t$: The set of players not on the TO's roster in week *t* and not on the roster of any opponent in week *t*: $FAPlayer_t = N - |TOPlayer_t \cup OppPlayer_t|$.

y_{ti}: A binary variable indicating whether player *i* is on the TO's roster during week *t*.

x_{ti}: The projected score of player *i* during week *t*. In week *t*, this figure is based on published projections from fantasy host websites; otherwise, it is the average projected weekly score based on season projections (i.e. P(*i*)/T).

f_{ti}: A binary variable indicating whether player *i* acts as a "flex" player for the fantasy team's starting lineup in week *t*.

a_{ti}: A binary variable indicating whether player *i* is included in the starting lineup of the team owner's fantasy team during week *t*; this value is found through the starting lineup optimization model.

WPosLimitL(j): The minimum number of players at position *j* that may be included in a starting lineup during any week.

WPosLimitU(j): The maximum number of players at position *j* that may be included in a starting lineup during any week.

5.5.2 Model Assumptions

1. Only one "add/drop" free agency move can be made per week *t*.

2. Opponent team owners do not make free agency moves.

5.6 The Free Agency Optimization Algorithm

The objective of the free agency optimization algorithm is to determine whether there is additional season-long "value added" to the TO's roster by adding a player in week *t* from the set

$FAPlayer_t$ and subsequently removing a corresponding player. If the addition of a player from $FAPlayer_t$ and subsequent removal of another player from $TOPlayer_t$ improves the season-long projected scoring for the team, then the "add/drop" move considering those two players should be executed.

5.6.1 The Free Agency Optimization Algorithm

At the beginning of each week *t*:

1. Create a "dummy" roster and set of free agents to test "add/drop" moves on: $TOPlayerD_t = TOPlayer_t$; $FAPlayerD_t = FAPlayer_t$.

2. For each player *i* in $FAPlayerD_t$:

   3. Add player *a* from $FAPlayerD_t$ to $TOPlayerD_t$ and move another player *d* from $TOPlayerD_t$ to $FAPlayerD_t$: $FAPlayerD_t = FAPlayerD_t - \{a\} + \{d\}$; $TOPlayerD_t = TOPlayerD_t + \{a\} - \{d\}$.

   4. Run the starting lineup optimization model with updated roster $TOPlayerD_t$.

   5. If the value of Z_3 is greater than the value of Z_3, consider adding player *a* to $TOPlayerD_t$ and dropping player *d* to $FAPlayerD_t$.

6. If no "add/drop" moves improve the value of Z_3 above the baseline value given by $TOPlayer_t$, make no moves in week *t*. If at least one "add/drop" move improves the value of Z_3 over the baseline value, make the roster move which yields the greatest value of Z_3:

   $TOPlayer_t = TOPlayer_t \setminus \{d\}$; $TOPlayer_t = TOPlayer_t \cup \{a\}$

   $FAPlayer_t = FAPlayer_t \setminus \{a\}$; $FAPlayer_t = FAPlayer_t \cup \{d\}$

Essentially, the free agency optimization algorithm tests whether adding a player *a* from the pool of free agents and subsequently dropping a player *d* from the team's roster improves the projected final score of the team's starting lineups in the remaining weeks of the season (from *t* through *T*). The player whose contributions yield the highest projected score from the team is added to the roster, regardless of the number of players in the free agent pool who may project to contribute to the team's final score total.

Chapter 6

Conclusion and Possible Extensions

This final chapter describes results and conclusions gleaned from the creation of multiple optimization models that may be used to improve decision making for fantasy football team owners at the initial draft and also during the in-season play period.

6.1 Conclusion

In this thesis, decisions in two separate periods of the fantasy football season were considered: the initial selection of players during the draft phase and the selection of players to include in the starting lineup and to add from free agency during the in-season phase.

In Chapter 3, the projected scores and actual scores of NFL players were compared, and the distributions of each statistic were analyzed. Most NFL positions showed the same general relationship between projected and actual scoring: a high number of players actually scored within the densest areas of the projected scores, and deviations could be seen most clearly along the tails of the actual scoring distributions. In other words, most NFL players score at about their projected level; a few, however, score substantially lower than their projections, and a few score substantially higher than their projections. As a rule of thumb, though, season-long projections prove to be accurate for the vast majority of NFL players.

Additionally, in Chapter 3, the week-to-week variance of NFL player performance was discussed. For nearly all positions, it was shown that the average weekly variance of players shows a right-skewed distribution. The majority of players fell in about the same area of average weekly variance for their positions, but some players certainly showed a substantially less consistent weekly scoring pattern.

The final substantial section of Chapter 3 investigated the tradeoff between the average number of points scored by an NFL player and the week-to-week variance of that player. For all positions, scatterplots with lines of best fit showed that players who scored highly also tended to have a higher weekly variance. Of course, this trend was more pronounced for certain players and less pronounced for others. This difference in how much each player followed the general rule of thumb of increasing variance with increasing performance was a main premise which guided the study of Chapter 4.

In Chapter 4, a bi-criteria optimization model was developed to help the team owner make decisions during the initial player selection phase of the fantasy league draft. This optimization model sought to maximize the projected season-long scores of players on the fantasy team while simultaneously minimizing the total variance of the players on the team. By creating an efficient frontier of fantasy teams, the tradeoff between projected scoring and total variance for a team was thoroughly investigated. While the projected scoring of teams did tend to decrease with more strict bounds on the maximum allowed team variance, substantial improvement was able to be made to teams, allowing them to score very highly and very consistently. Generally speaking, teams with higher variances could decrease their variances substantially while only decreasing their projected total points marginally. Teams with lower variances, though, could not decrease their variance further without substantially decreasing their season-long scoring projection.

To make the draft strategy model more robust to the behaviors of opponent team owners, an algorithm was developed which allowed for the simultaneous maximization of projected points and minimization of team variance at each team owner draft pick. Modeling the draft in this way allowed for the optimization model to re-assess the players available at each draft pick and select the best player to satisfy both optimization criteria while also considering the past and likely future behaviors of opponent owners as predicted by player rankings.

In Chapter 5, an optimization strategy for in-season roster management was developed. First, a maximization model was created to determine the optimal starting lineups for each week of the season given the players already on the team owner's roster. This model was then used as part of an algorithm to assess the season-long impact of free agency moves on the team's final projected score. By considering the season-long ramifications of immediate roster changes, the team owner would be more likely to make moves in free agency which would substantially improve the scoring projections of their team. By using the free agency optimization algorithm, the team owner could assess at the beginning of each week whether any player additions from free agency or player subtractions from their own team would improve their scoring outlook for the rest of the season. With this information, they could make more informed decisions which are only likely to improve their team's performance.

The use of statistics and analytics within fantasy sports is certainly not a new endeavor, but this thesis presents novel techniques for modeling the draft phase of the season as well as the season-long impact of immediate roster changes made during the year. This thesis is also the first attempt to marry portfolio optimization techniques with fantasy football, essentially viewing each player as an asset with a risk and expect reward during the draft and using techniques similar to those of Markowitz to find optimal teams with risk levels that are palatable to the team owner. Additionally, this thesis presents a useful method for modeling and assessing the season-long impact of roster changes made during the season, allowing team owners to make only in-season free agency transactions with behoove their team and improve the team scoring projection. This thesis has made substantial progress in applying optimization techniques to the entire realm of season-long fantasy football team management.

6.2 Possible Research Extensions

While this analysis has made substantial progress in applying advanced statistical modeling techniques to fantasy football team management, there are certainly avenues that this analysis leaves unexplored. Further research can certainly be carried out in a number of areas.

6.2.1 Model Tractability

In early rounds of the initial draft (i.e. when n_k is low), there are many players to consider adding to the TO's fantasy team roster. In these early cases specifically, the bi-criteria optimization model can take several minutes to produce results. In typical fantasy drafts, the time that a team owner has to select a player during the draft may be between two and three minutes in these early rounds and even shorter times during later rounds. If the draft optimization strategy is to be made widely available and intended to be used by many, some heuristics should be applied to make the draft optimization more tractable and more easily solvable.

In addition to the draft optimization model, the free agency optimization algorithm should be made more tractable as well. For many fantasy host sites like ESPN and Yahoo!, the player universe N includes nearly 1,000 players. While the majority of these players are not relevant and are typically projected to score 0 points each week, they are still included in the full set of players. With that dynamic in place, the free agency optimization algorithm is meant to test the possibility of adding each player to the team owner's team, but this is clearly unrealistic with so many players to consider. Frankly, though, the consideration of every player is also not necessary—players who are consistently projected to score 0 points or who are not even on NFL teams are likely not worth considering adding to the team owner's roster. With 160 players included on the rosters of a 10-team fantasy league, perhaps only the top 225 or 250 players should be considered for addition from free agency. The method by which the considered player pool is narrowed down and the actual number of players considered is another area of potential future research, though, which could greatly improve the tractability of the model and feasibility of use of the free agency algorithm.

6.2.2 Inclusion of Variance in In-Season Optimization

The draft optimization model seeks to maximize the total number of points scored by drafted players while minimizing the week-to-week variance of those players. This bi-criteria optimization model is used because teams that score a high number of points on a consistent basis are more likely to finish in the top few spots of the standings of a fantasy football league than inconsistent teams or teams that consistently score a low number of points.

In some contrast to the draft optimization model, though, the starting lineup optimization model does not take week-to-week variance into account in considering which players to include in the starting lineup. It's certainly possible that the model could choose the players with the highest variances to include in that lineup, as we have already seen a trend that the variance of a player typically increases with the scoring of a player. Placing a constraint on the variance of the starting lineup could be a worthwhile future investigation.

Additionally, the free agency optimization algorithm does not take the week-to-week variance into account in considering adding or dropping players. This means that over the course of the season, a drafted team with a high expected number of points and low weekly variance could be transformed into a team with an even higher expected number of points but now also high weekly variance. There is some merit in having “boom or bust” players on a fantasy football team; these players just require that the team owner choose to add them to the starting lineup for weeks in which they’re likely to “boom,” or score a very high number of points. The fact that there is no consideration for variance in player performance in the free agency optimization algorithm, though, is admittedly a shortcoming of the model and something that should be explored as a future point of research.

6.2.3 Consideration of Multiple Free Agency Moves in Each Week

The free agency optimization algorithm dictates that the player in the free agency pool which improves the expect total points scored by the team the most should be added to the team roster. In reality, team owners are typically allowed to make as many free agency additions as they would like during each week. However, considering the possibility of multiple free agency transactions was simply not realistic given the tractability concerns of the free agency optimization algorithm.

In the future, researchers may want to explore the possibility of adding and dropping multiple players during a given week *t*. This extension would better reflect the real-world possibilities of roster transactions in fantasy football leagues. In order to consider these types of moves, though, the starting lineup optimization model and free agency optimization model would need to be made more tractable.

6.2.4 Free Agency Logic and Waiver Wire Claims

In many cases, multiple fantasy team owners may want to claim the same player from the free agent pool at the same time. In order to fairly distribute these desired players to the different team owners, fantasy contest host sites implement some sort of logic to restrict the number or frequency of moves of team owners.

In some cases, fantasy football host sites may dictate that the team which scored the fewest points in week *t-1* has first priority in choosing which, if any, players to add to their team from the free agency pool in week *t*. In other cases, host sites choose to allow the team owner whose team is lowest in the league standings at the beginning of week *t* to have first priority in choosing free agency additions. Other sites even provide each team with a budget of some amount of money for the entire season and allow team owners to anonymously bid on free agent players; in this setup, at the end of the day, the fantasy team owner who bid the highest amount for a free agent is awarded the player for their team roster.

Likely the most used free agency priority logic is "waiver wire" claims. Immediately after the draft, the priority for adding free agents is set in reverse draft order (e.g. the player who picked last in the first round of the draft has the highest priority in adding free agents). Each week, team owners have the choice of whether or not to submit a claim for a free agent player. If the team owner does submit a claim and is awarded the player, they lose their place in the priority order and are moved to the very bottom of the priority lost.

In many of these free agent priority systems, there is a cost associated with adding a free agent to a team owner's team. In the case of bidding for players, that cost is obvious—whatever sum of money the team owner spends on a free agent is not available to spend on free agents in future weeks. In the case of waiver wire claims, though, that cost is less clear. The cost of choosing to add a free agent in week t is the opportunity cost of no longer having that same priority to choose to add a free agent in subsequent weeks. Quantifying this phenomenon and incorporating it into the logic of whether or not a team should choose to add a free agent is a logical and worthwhile future area of study.

6.2.5 Other Methods for Considering Free Agency Decisions

In this study, we assumed a somewhat "rational owner," meaning that the team owner always sought to improve the total number of points scored by players in their team's starting lineup. Sometimes, though, team owners may monitor free agents and players on their own rosters and make more speculative decisions.

Consider the instance of a player on the team owner's roster who consistently underperforms his projected weekly score—the NFL player is simply not playing well. Perhaps there are even rumors that the player could have his role reduced on his NFL team or could be released by his NFL team entirely due to poor performance. At the same time, consider a player in the fantasy free agent pool who has consistently overperformed his projected weekly score. Perhaps this player still does not score as much as the already-rostered player, but they seem to be in line for more consideration on their NFL team and more opportunities to score in NFL games. All things considered, the fantasy owner sees that the value of their currently owned asset is trending down, and the value of a player available to them—the free agent—is trending up. The team owner may believe that in a few weeks, the current free agent's performance may surpass that of the player currently on the roster. Of course, this free agent is available to all team owners, and each of them sees the same improving performance. The team owner in question must decide whether the likelihood of the free agent surpassing the player currently on the roster in a few weeks is enough of an incentive to add that free agent and drop the currently rostered player.

Additionally, NFL players become injured often, and the severity of their injuries can play into their projected weekly scoring for the rest of the season. If a player on a team owner's roster is going to miss four games, they must consider whether it is more valuable to keep that player on the roster in a bench position or to drop the player to free agency so that they can pick up another player currently in the free agent pool. Even though this currently free agent may be an inferior player compared to the one already on the roster, the injured player requires a space on the team's bench, taking up an opportunity for the team owner to hold other players who at least have the possibility of contributing points to the team.

Probabilistic modeling or more advanced dynamic programming techniques could be explored in an effort to consider more complex free agency decisions.

6.2.6 In-Season Trades and Trade Optimization Modeling

Adding players to the team owner's roster is not the only avenue of changing the players on the roster. Team owners may choose to swap players between their fantasy teams. As long as both owners agree to the terms of the trade, they are able to swap whichever players they'd like. Without other human team owners, this logic is not easily implemented into a season-long simulation. Considering team needs, owner preferences for certain players, and other factors seems nearly impossible to adequately and easily model. However, this of course is an area which can be explored in the future.

Additionally, most other team owners do not make data-driven decisions based on optimization techniques, algorithms, and modeling like the ones outlined in this thesis. Creating an algorithm with dynamic programming which could weigh the value of players in a trade could be invaluable for a team owner. If another owner proposed to trade a set of players in exchange for a typical starter or two, the team owner could use a dynamic programming approach to determine the season-long impact of making that trade. If the player swap substantially improves the projected final point total for their team, the owner should likely accept the trade; if the swap does not improve their final projections or only does so meagerly, the owner may want to decline the trade. Assessing all trades in this way would give the team owner the tools to find trade opportunities to greatly improve the projection of their team.

6.2.7 Value of Roster Depth

The draft optimization model seeks to maximize the total number of points scored by all players on the team owner's roster while minimizing the week-to-week variance of those players. In somewhat of a contrast, though, the free agency optimization algorithm seeks to maximize the total number of points scored by players in the starting lineup of the owner's roster. Obviously, only players in the starting lineup are able to accrue points each week, but having players on a team's bench who can easily move into the starting lineup in the event that a projected starter becomes injured or underperforms can be invaluable. The question of roster depth comes down to resource allocation—how does a team owner want to spend their draft picks, and how do they want to spend their priority in picking up free agents? Finding a way to value the depth of players at certain positions on a roster could be another way to improve upon the modeling techniques used to ultimately enhance the chance of winning a fantasy football league.

6.2.8 Weight of Playoff Games

Depending upon the number of teams in a fantasy football league, the playoffs for the league take place during the final few weeks of the NFL regular season. In the case of a 10-team league, the regular season is typically the first 13 weeks of the NFL regular season, with the playoffs beginning in week 14 and continuing through week 16. These dates can change based on the preferences and settings of the league.

In any event, the performance of NFL players in fantasy playoff games is clearly more impactful in terms of probability of winning a fantasy football league than the performance of NFL players in regular season games. As the free agency optimization algorithm considers adding players and dropping others, it does not place any additional weight on any player’s projected performance during the playoffs as opposed to their performance during the regular season. Though this effect could be difficult to quantify, adding some term to the model which prioritizes scoring during playoff weeks could be a valuable and helpful tool for team owners looking to maximize the chance of winning their fantasy football league.

6.2.9 League Pay-Out Structures

Each fantasy football league “pays out” differently. For example, in a 10-team league, each player may contribute $50 to the pot of money to be split up based on the final results of the league. Some fantasy leagues may decide to award the entire $500 prize to the winner of the league, others may decide to employ a tiered system and split the $500 among the first, second, and third place finishers, and others still may decide to award some amount of money to the team with the best regular season record and the rest of the money to the best playoff teams.

Throughout this study, the objective has been to maximize the total number of points scored by players on a fantasy team—specifically in the starting lineup—because that is the best way to improve chances of general success for a fantasy team. To this point, no league pay-out structures have been considered. If a team owner’s objective, though, is not to score the highest number of points consistently—in other words, if a team owner’s objective is instead to maximize winnings based on the league pay-out structure—then some of the modeling presented in this paper should be changed. For instance, in a “winner take all” league, perhaps the team owner should have a higher tolerance for risk in their team players, not prioritizing the minimization of week-to-week variance of the players that they draft. In contrast, if the league pay-out structure awards money to the top four or five teams out of ten, the team owner would likely value consistent play, as it would lead to finishes in the top few places more often than not. Adding some sort of consideration of the league’s pay-out structure—likely by weighting different aspects of models—would be valuable in allowing the optimization techniques to serve another objective.

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